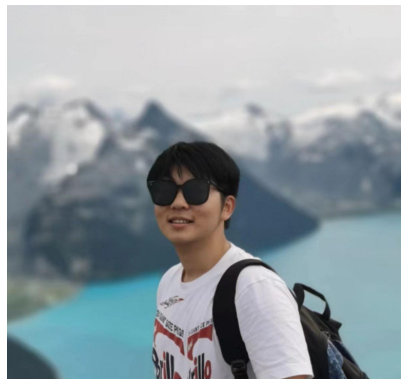
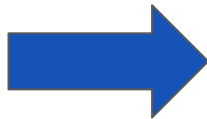


Welcome to the BabyVLM Tutorial!

Please visit our tutorial webpage for our schedule, slides, and data explorer!



Wenqi Wang



Max Whitton



Boqing Gong



Michael Frank



Shawn Wang

BabyVLM-v2: Opening and Motivation

Toward **Developmentally Plausible** Vision Foundation Models



BabyVLM-V2

Boqing Gong

bgong@bu.edu

Presented by

Max Whitton: maxwh@bu.edu

What's wrong with LLMs in computer vision?

How to fix it? BabyVLM-V2 etc.

What does BabyVLM unlock?

Making Sense of Touch from the Child's View for Contrastive Learning

Max Whitton, Zecheng Wang*, Puchen Liu*, Quang Tuan Truong, Shengao Wang, Manaswi Yadamreddy, Oktay Ozel, Visista Jayanti, Saniya Sekhon, Hanna Samuel Tadesse, Lawrence Miao, Junjie Wang, Jiasen Lu, Chen Yu¹, and Boqing Gong

POSTER

Sun, Jun 7, 2026 · 2:30 PM – 4:30 PM PDT

ExHall A 219

The LLM Bottleneck: Why Open-Source Vision LLMs Struggle with Hierarchical Visual Recognition

Yuwen Tan · Yuan Qing · Boqing Gong

POSTER

Sat, Jun 6, 2026 · 3:45 PM – 5:45 PM PDT

ExHall A & F 211



BabyVLM-V2: Toward Developmentally Grounded Pretraining and Benchmarking of Vision Foundation Models

Shengao Wang · Wenqi Wang · Zecheng Wang · Max Whitton · Michael Wakeham · Arjun Chandra · Joey Huang · Pengyue Zhu · Helen Chen · David Li · Jeffrey Li · Shawn Li · Andrew Zagula · Amy Zhao · Andrew Zhu · Sayaka Nakamura · Yuki Yamamoto · Jerry Yokono · Aaron Mueller · Bryan A. Plummer · Kate Saenko · Venkatesh Saligrama · Boqing Gong

What's R in CVPR? Do we still need P?

TRADITIONAL DEFINITION

1. CVPR = Computer Vision & Pattern **Recognition**

EMERGING PARADIGM

2. CVR = Computer Vision & **Reasoning**

Question: Are *Large Language Models (LLMs)* the key to this reasoning?

What are vision foundation models?

AI models pretrained on **large visual datasets** to support **many tasks**

General-purpose models

A new paradigm for computer vision and neuroscience research

Families of vision foundation models

Generative VLM

LLaVA / Qwen-VL



“What is happening?”



“They are high-fiving.”

Image generation

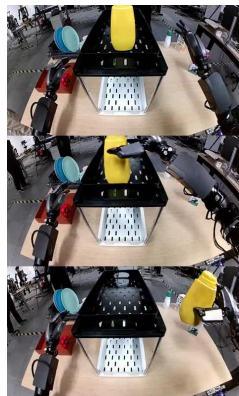
Stable Diffusion

“An astronaut playing piano in space”



World models

Cosmos-Predict2



Earlier

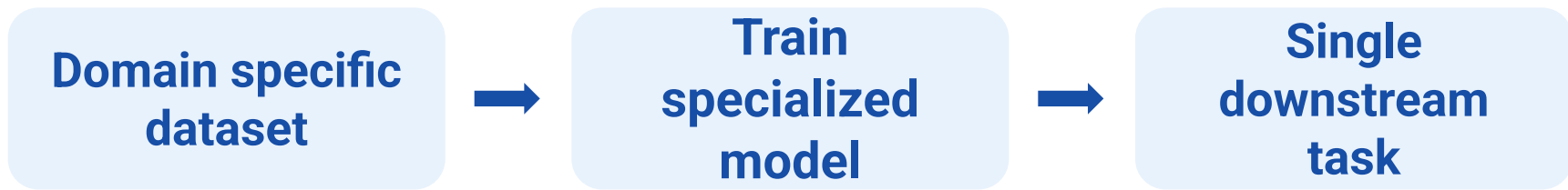


Later

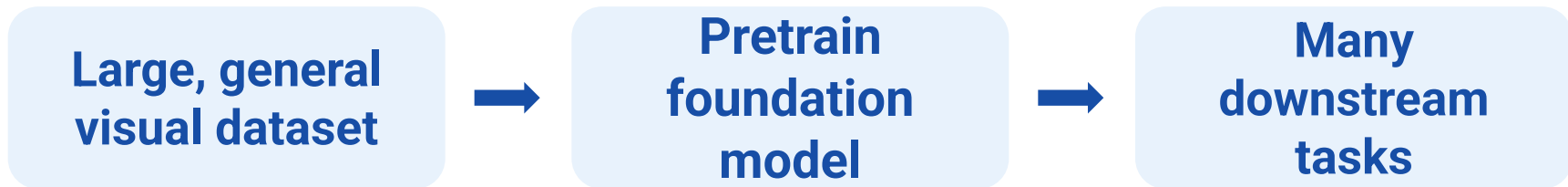
Observation + condition
→ possible future video

What's new with vision foundation models?

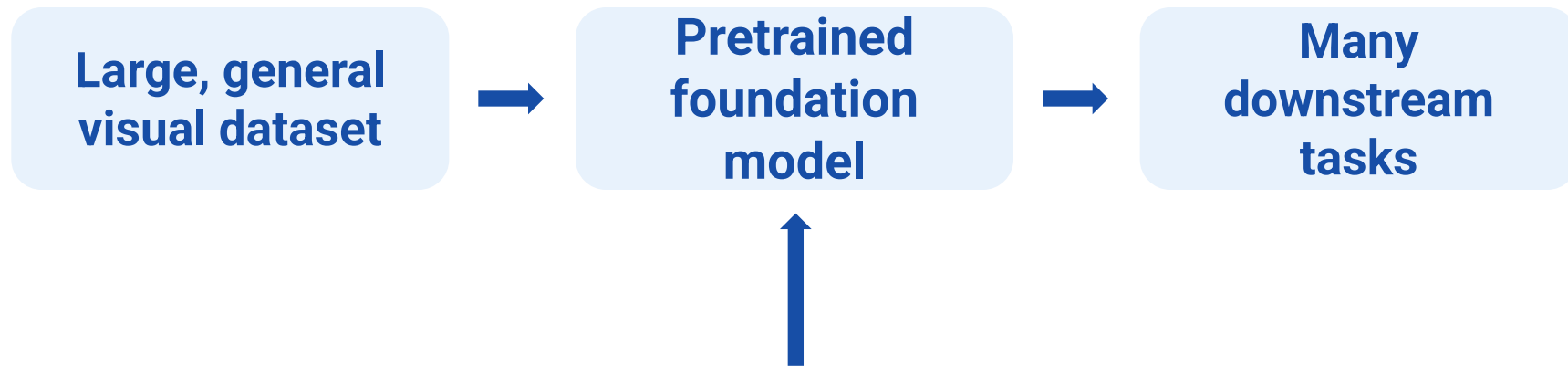
Old paradigm:



New paradigm:



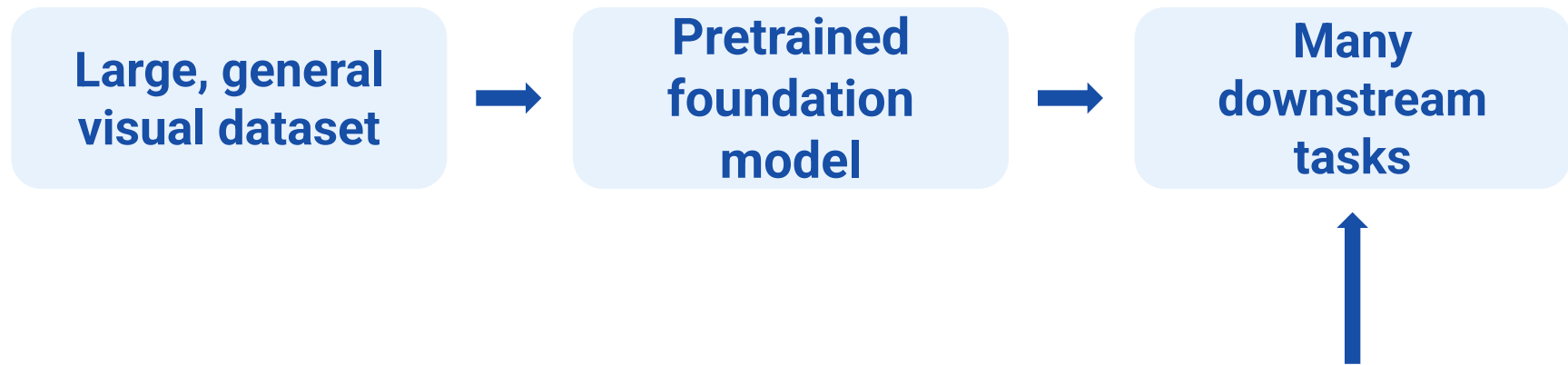
The problem with the new paradigm



This step is prohibitively expensive

Often concentrated in well-funded industry labs

The problem with the new paradigm



Researchers are forced to work here

Open source VFMs: **post-training**

Closed source VFMs: **"API research"**

Models can also align, represent and reconstruct

Different model families support different visual tasks

Image-text alignment

CLIP



"A dog"



Nearby features in a
shared representation

Match images and text

Visual representations

DINO

Image

Attention map

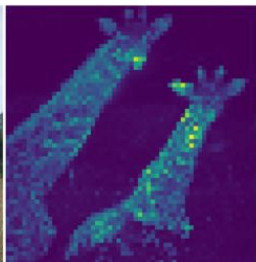


Image $\rightarrow$ visual features
Useful across visual tasks

3D reconstruction

VGGT

32 Views



0.51 s



Scene views $\rightarrow$ 3D geometry
Cameras, depth, points

Why pretraining in academia?

Who gets to build the starting point?

Pretraining

Often concentrated in
well-funded industry
labs

Post-training

Academia adapts
models
others have already
built

API research

Academia studies
models
through limited
interfaces

Academic pretraining lets us test how models learn from scratch.

The LLM Bottleneck



SCAN ME

Why Open-Source Vision LLMs Struggle with Hierarchical Visual Recognition

Yuwen Tan · Yuan Qing · Boqing Gong

Question: Identify the taxonomic group of the animal in the image. Answer with the letter of your chosen option.



Query Image

Level 1:

A. Vertebrate B. Invertebrate

Level 2:

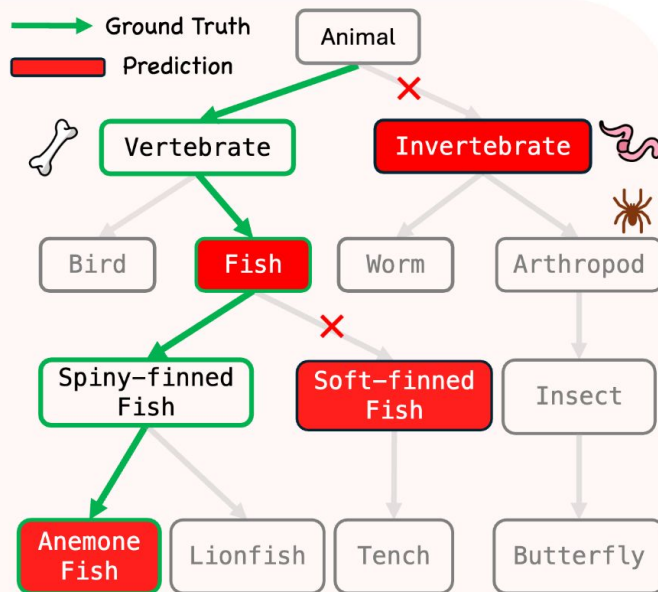
A. Arthropod B. Fish
C. Bird D. Worm

Level 3:

A. Spiny-finned Fish
B. Soft-finned Fish
C. Sea Slug
D. Sea Anemone

Level 4:

A. Lionfish B. Tench
C. Anemone Fish D. Butterfly



LLMs are the bottleneck for visual taxonomy



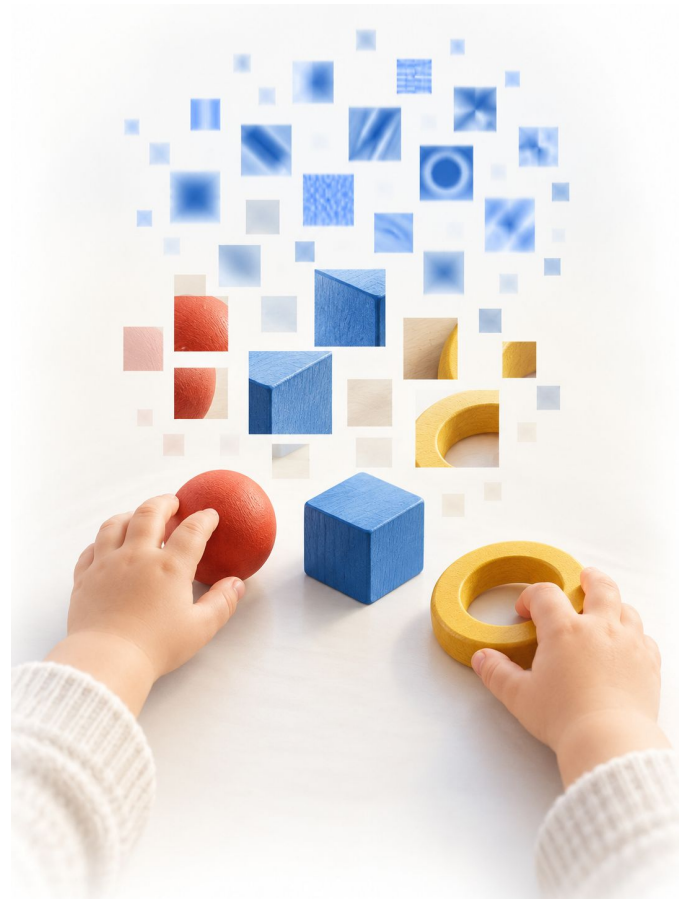
Visual data retains useful information.

Language only models lack taxonomy knowledge,
even after visual instruction tuning.

Pretrain our own vision-first LLMs

LLMs are still remarkable.

Can we ground their learning
in visual experience?



Progress needs a broader community

Public trust

Academic pretraining lets us test how models learn from scratch.

Understanding this makes scrutiny possible.

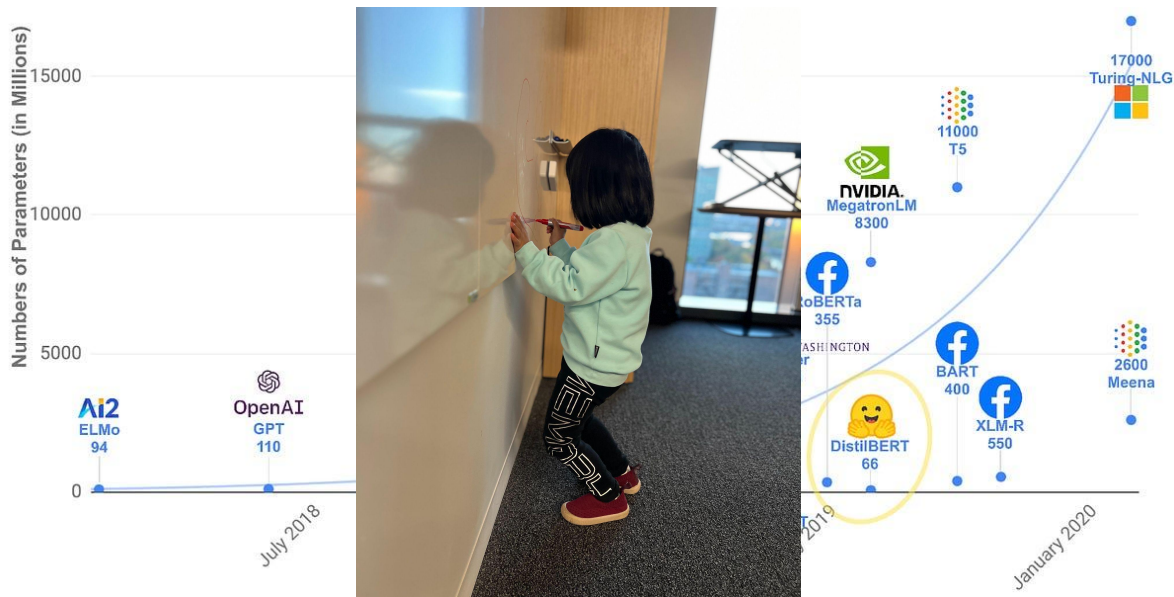
New ideas

A wider research community can explore what comes next.



Two proven paths to learning in foundation models

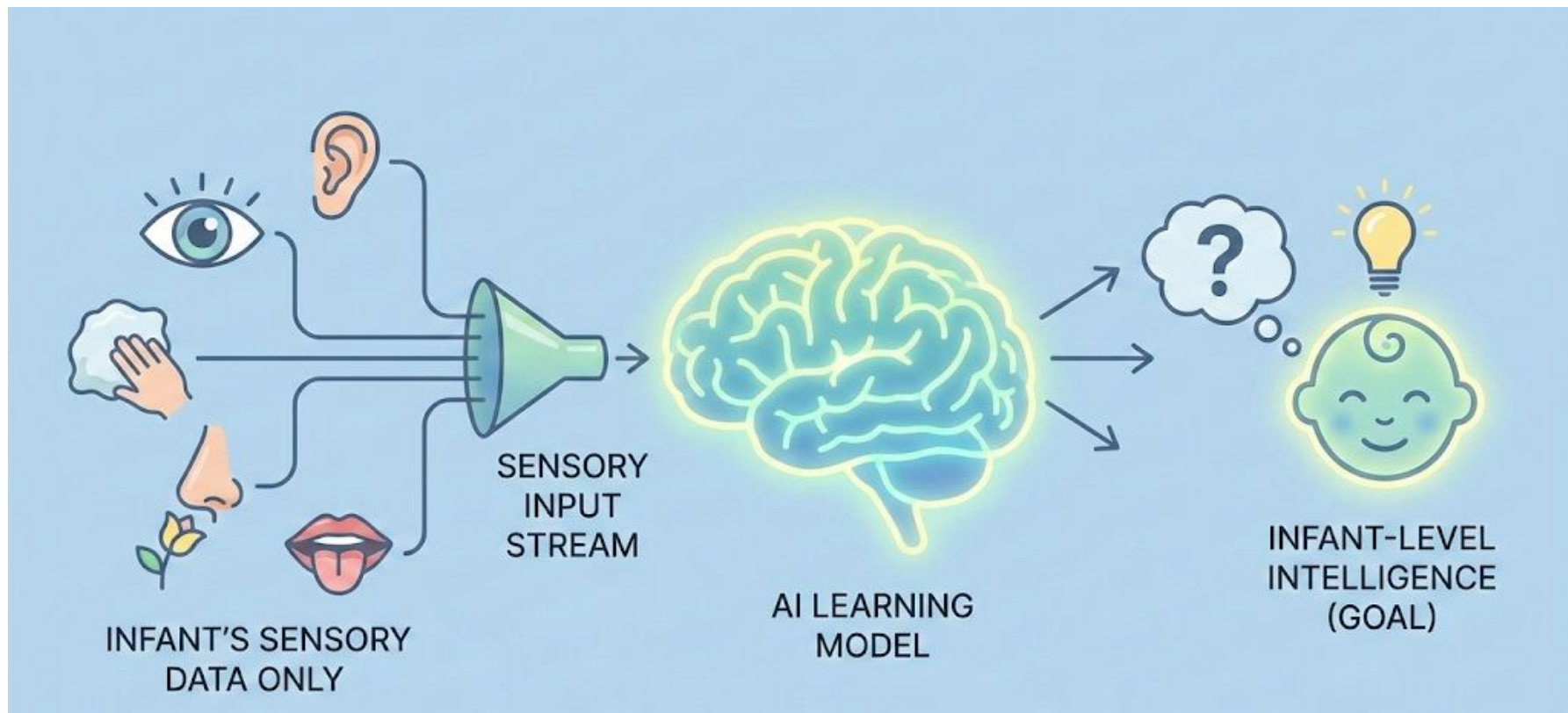
Human Scaling Development



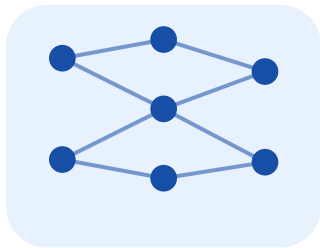
Artificial 3-year-old intelligence



Can we achieve infant-level intelligence using infants' sensory data?

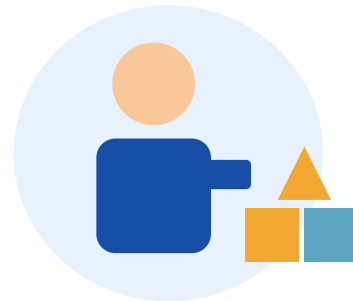


A bridge between vision and human development



Computer vision

Leverage existing learning theory to improve models.



Developmental psychology

Test ideas about learning from sensory experience.



Can we achieve **infant-level intelligence** using infants' sensory data?

POSTER

📅 Sat, Jun 6, 2026 · 3:45 PM – 5:45 PM PDT

📍 ExHall A & F 211



BabyVLM-V2: Toward Developmentally Grounded Pretraining and Benchmarking of Vision Foundation Models

Shengao Wang · Wenqi Wang · Zecheng Wang · Max Whitton · Michael Wakeham · Arjun Chandra · Joey Huang · Pengyue Zhu · Helen Chen · David Li · Jeffrey Li · Shawn Li · Andrew Zagula · Amy Zhao · Andrew Zhu · Sayaka Nakamura · Yuki Yamamoto · Jerry Yokono · Aaron Mueller · Bryan A. Plummer · Kate Saenko · Venkatesh Saligrama · Boqing Gong





LLMs are slowing down computer vision research.

They **don't recognize** well, and yet they **reason**.



Can we achieve infant-level intelligence using infants' sensory data?

Why should we care?

Scientifically reasonable

Computationally friendly

Safety, novelty, benefiting AGI, ...

Intriguing: Babies!



VideoPrism video corpus:
>100 years total duration



A three-year-old human:
< 3 years of total wake time



Can we achieve infant-level intelligence using infants' sensory data?

SAYCam: A Large, Longitudinal Audiovisual Dataset Recorded From the Infant's Perspective

Jessica Sullivan¹, Michelle Mei¹, Andrew Perfors², Erica Wojcik¹, and Michael C. Frank³



The BabyView dataset: High-resolution egocentric videos of infants' young children's everyday experiences

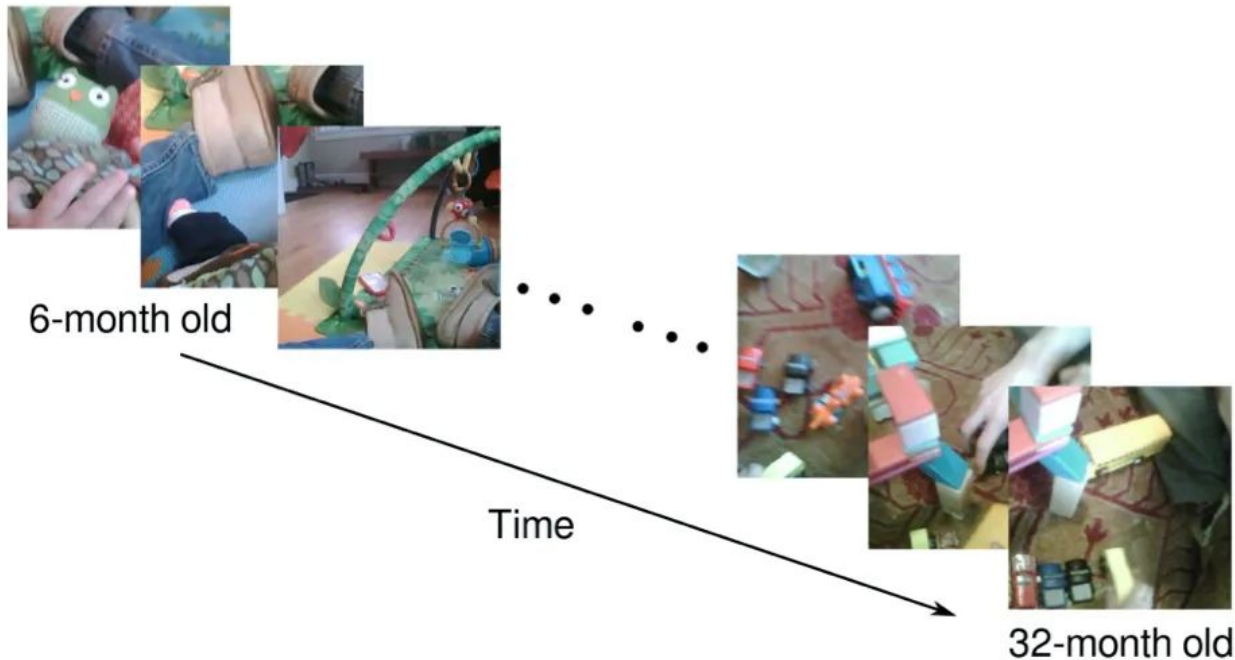


Bria Long^{1,2*} Robert Z. Sparks^{1*} Violet Xiang^{1*} Stefan Stojanov^{1*} Zi Yin¹
Grace E. Keene¹ Alvin W. M. Tan¹ Steven Y. Feng¹ Auddithio Nag¹ Chengxu Zhuang¹
Virginia A. Marchman¹ Daniel L. K. Yamins¹ Michael C. Frank¹



Can we achieve infant-level intelligence using infants' sensory data?

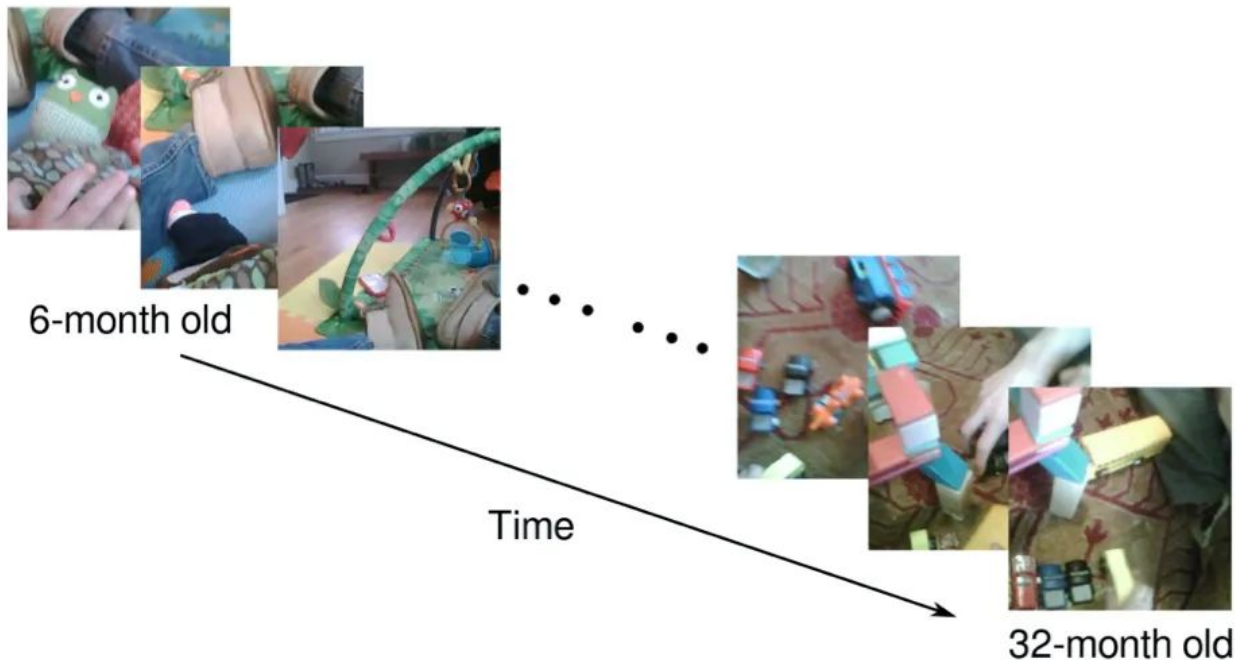
Longitudinal





Can we achieve infant-level intelligence using infants' sensory data?

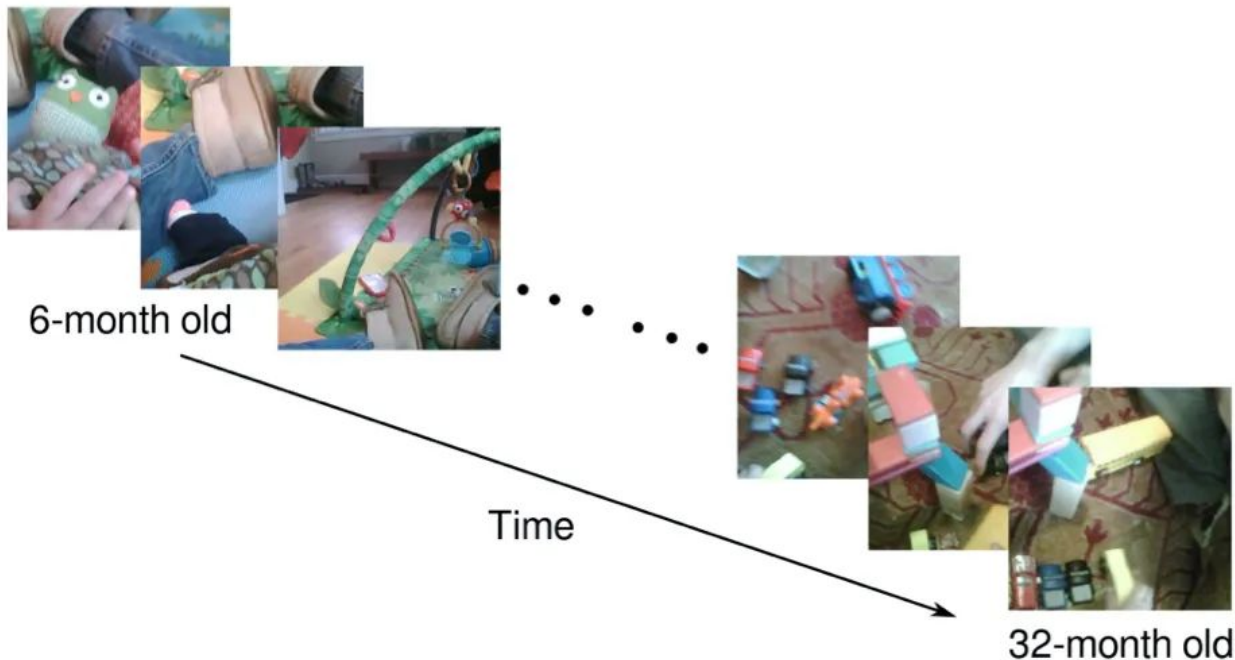
Longitudinal, egocentric





Can we achieve infant-level intelligence using infants' sensory data?

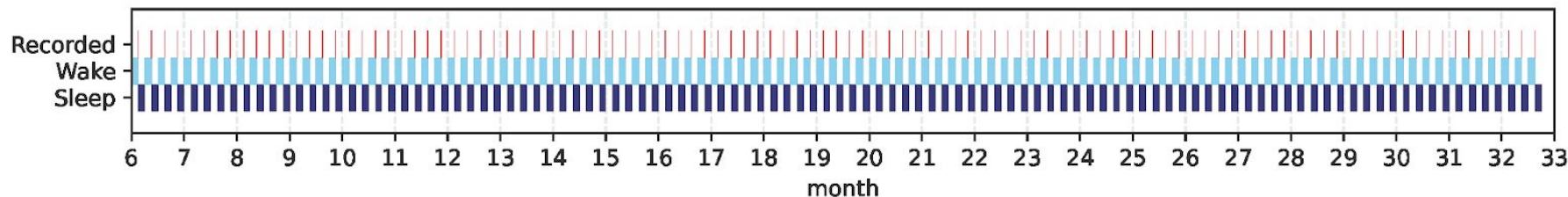
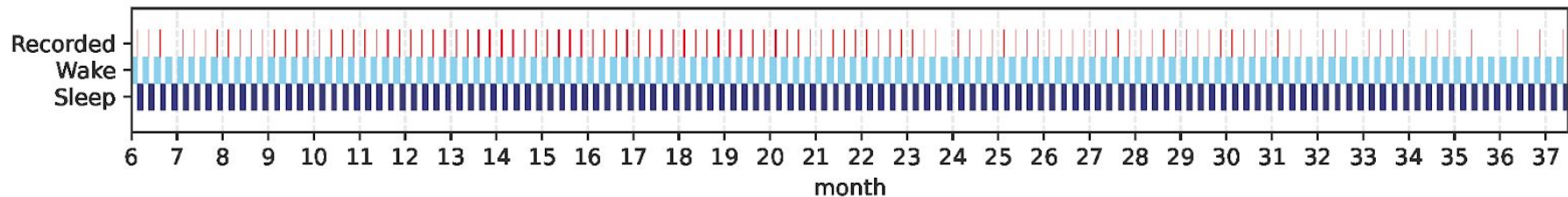
Longitudinal, egocentric video recording of what infants see and hear





Can we achieve infant-level intelligence using infants' sensory data?

Longitudinal, egocentric video recording of what infants see and hear





Can we achieve infant-level intelligence using infants' sensory data?

Longitudinal, egocentric video recording of what infants see and hear



Time: 14:15

Utterance: I'm going to sit over here because I'm going to cut some walnuts.



Time: 15:22

Utterance: It's really good with yogurt.



Time: 17:03

Utterance: Want to try a little piece of this?



Time: 18:46

Utterance: How about another blueberry?



Time: 18:59

Utterance: Oh that's a very sweet one.



Time: 00:38

Utterance: Can you do the white one?



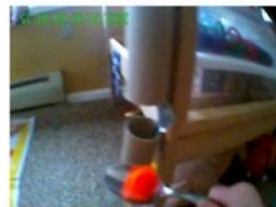
Time: 00:40

Utterance: You won't?



Time: 00:45

Utterance: Which one are you choosing?



Time: 00:48

Utterance: Hard, huh?



Time: 00:50

Utterance: Don't use your hand.

What is a vision foundation model?

Different model families in different use cases

Generative VLM

LLaVA / Qwen-VL



“What is happening?”



“They are high-fiving.”

Image generation

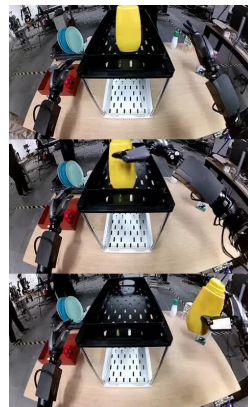
Stable Diffusion

“An astronaut playing piano in space”



World models

Cosmos-Predict2



Earlier



Later

Observation + condition
→ possible future video

Running example: generative vision-language models

Images and instructions in. Text out.

Image + instruction



"Describe this image."

**Vision-language
model (VLM)**

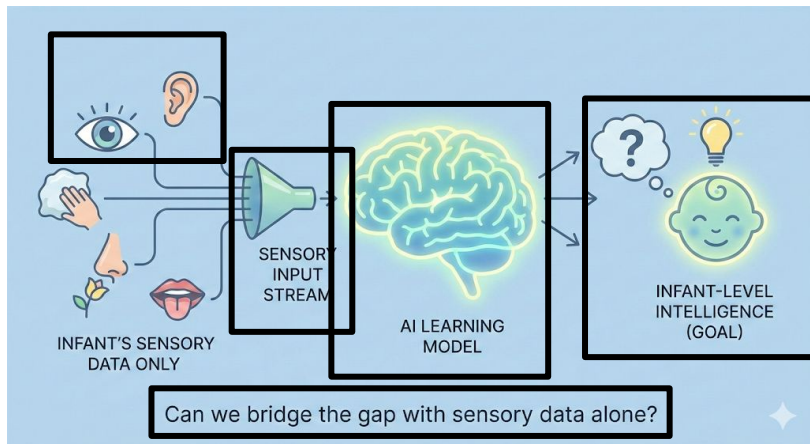
e.g., LLaVA or Qwen-VL

Text response

"A dog and a woman
are giving each other
a high-five on a beach."

Describe scenes, answer questions and read text.
Increasingly, these models also tackle visual reasoning.

Tutorial overview



1 Developmental data collection

***coffee break**

2 Pretraining data curation

3 Modeling and algorithm

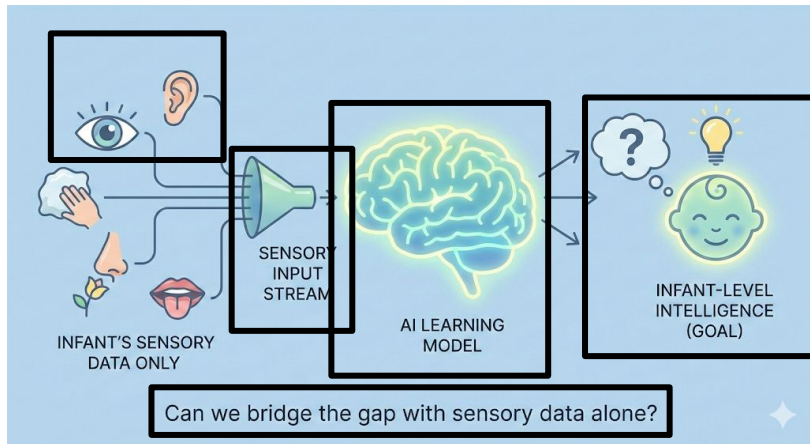
4 Developmentally aligned benchmarks

5 Instruction tuning, evaluation and results

6 Future work and Q&A

The BabyVLM Challenge

Briefly introduce the challenge



Break for talk from Michael Frank

10:00-11:00

<https://meet.google.com/cjr-jkjd-rpd>

Coffee break

11:00-11:20

BabyVLM-v2: Pretraining Dataset

Toward **Developmentally Plausible** Vision Foundation Models



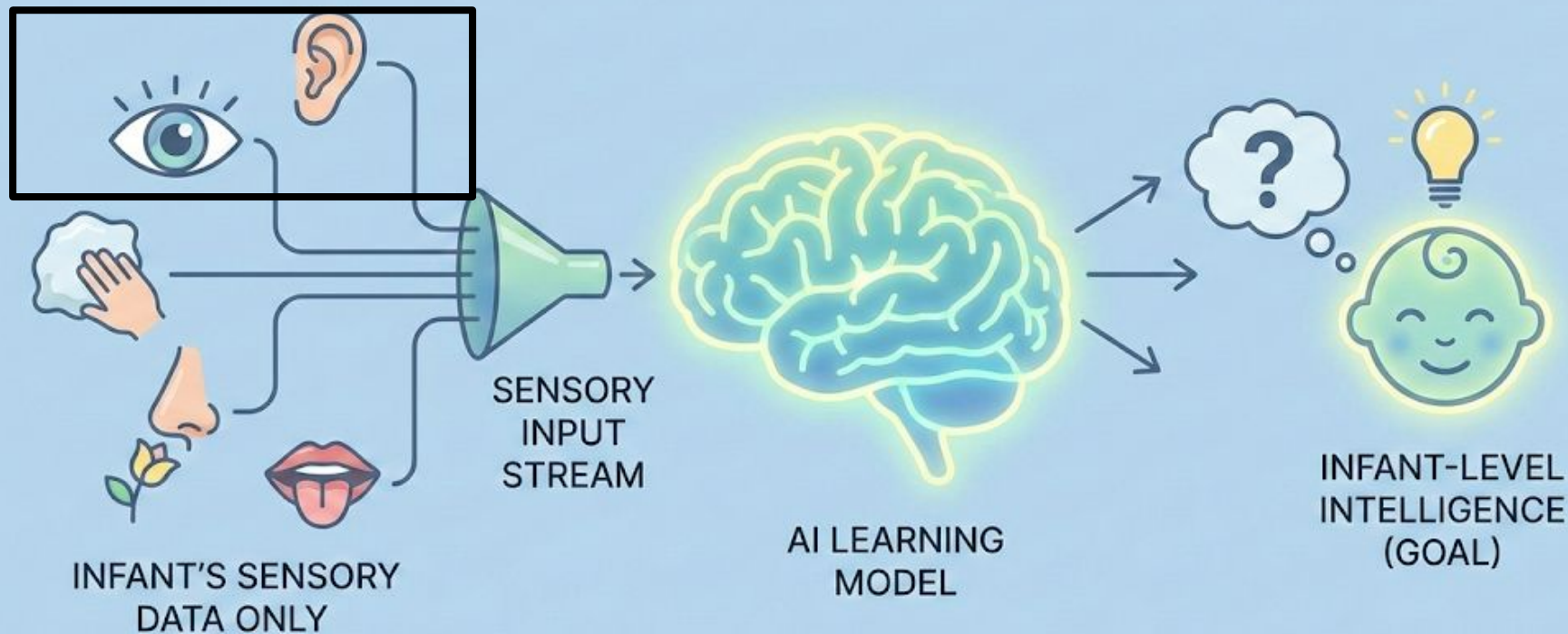
BabyVLM-V2

Wenqi Wang

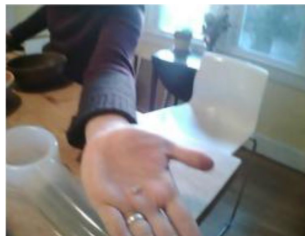
wqwang@bu.edu

Can we achieve infant level intelligence using infant's sensory data?

How to process raw sensory data?



SAYCam dataset



What is the recording frequency?

What objects do babies see?

Where do babies stay?

What do babies hear?

Recorded experience

3 infants

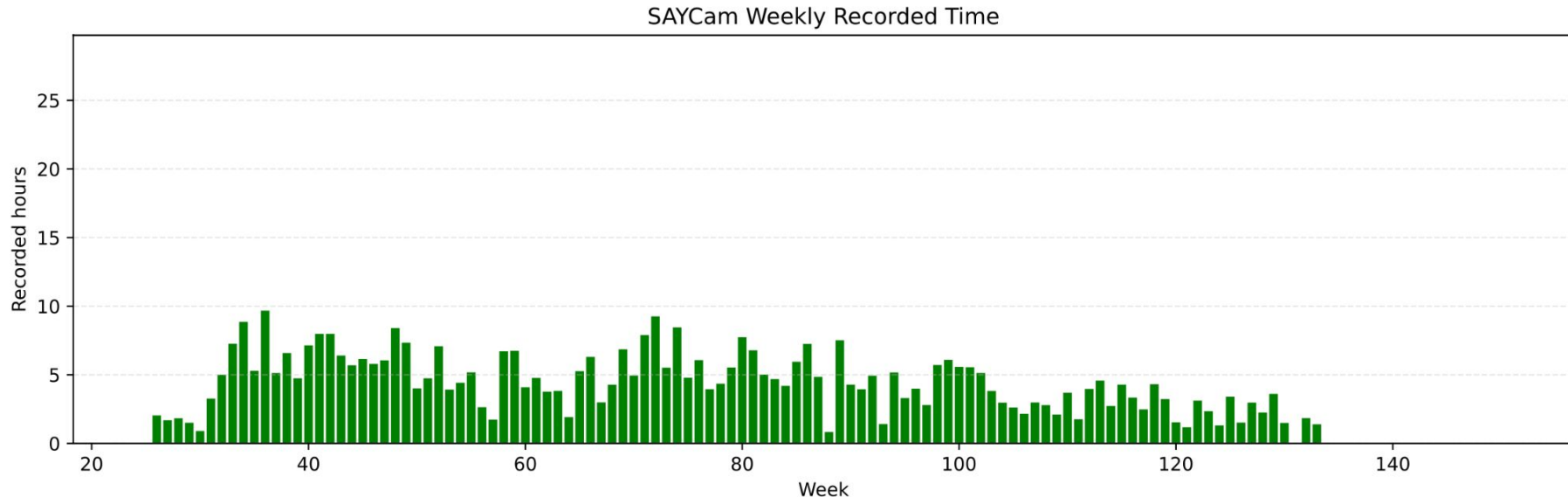
478 hours

~6–32 months

SAYCam dataset - statistical analysis

Recording frequency:

- ~4.5 hours / week
- distribute smoothly from 6 months - 32 months



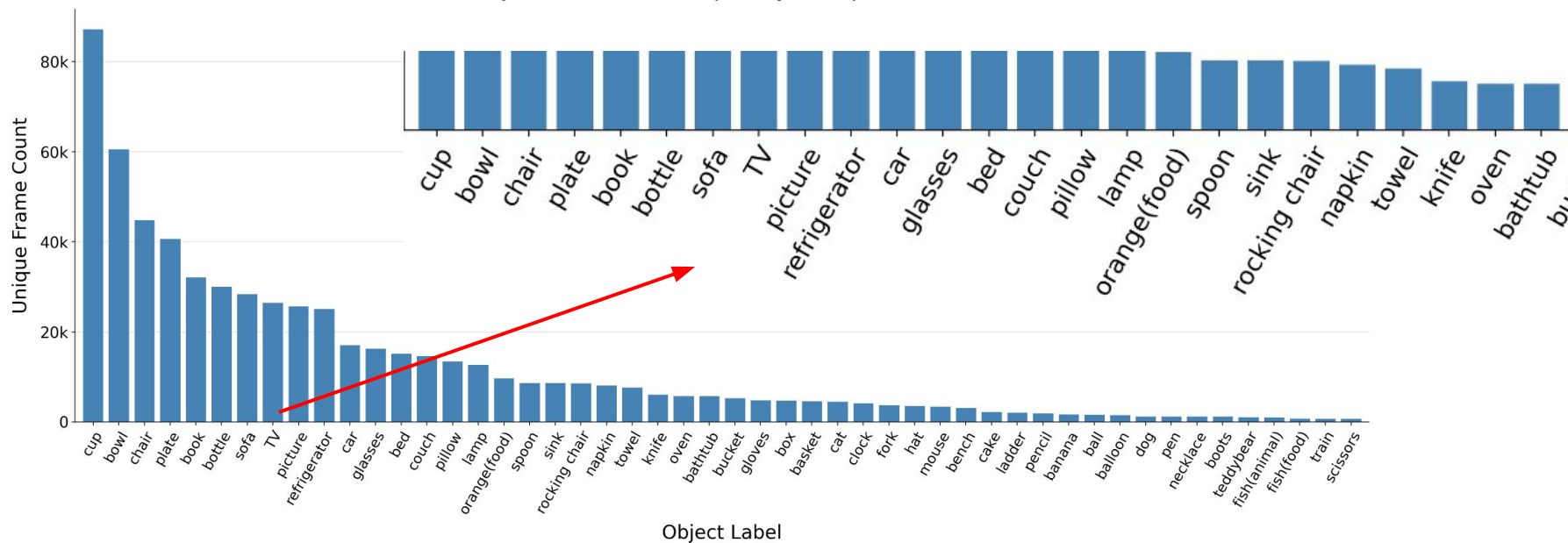
SAYCam dataset - statistical analysis

What do babies see and stay? Object detection

- Detection: 1 fps using YOLO8x-Worldv2
- Vocabulary: nouns from MB-CDI world bank (N=267)
- confidence > 0.65



SAYCam — Object Detection Frequency — Top 50 (threshold=0.65)



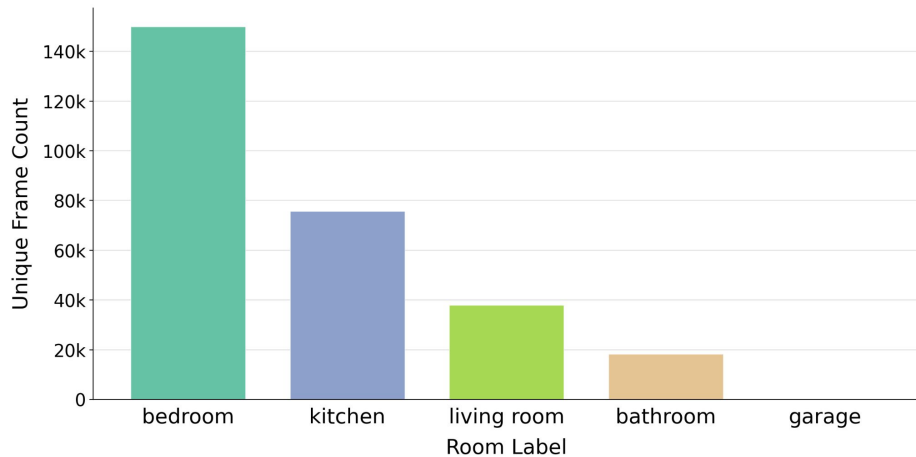
SAYCam dataset - statistical analysis

What do babies see and stay? Object detection

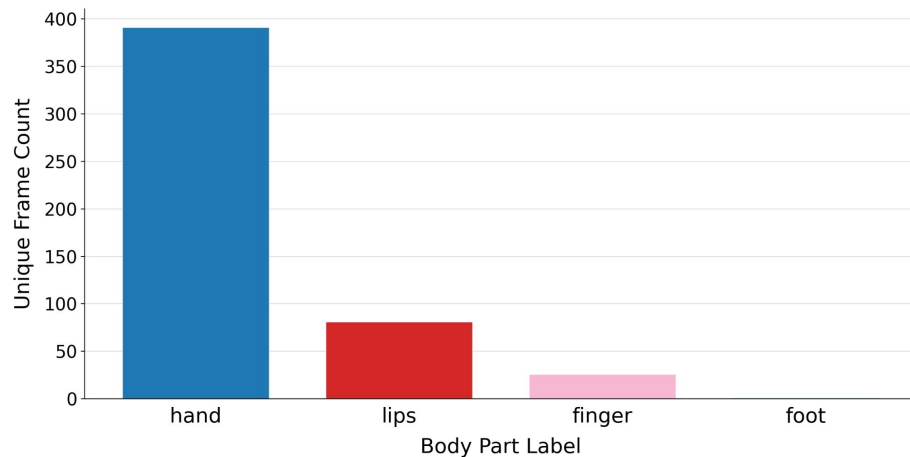
- Detection: 1 fps using YOLO8x-Worldv2
- Vocabulary: nouns from MB-CDI world bank (N=267)
- confidence > 0.65



SAYCam — Rooms at Home — Detection Frequency (threshold=0.65)



SAYCam — Body Parts — Detection Frequency (threshold=0.65)

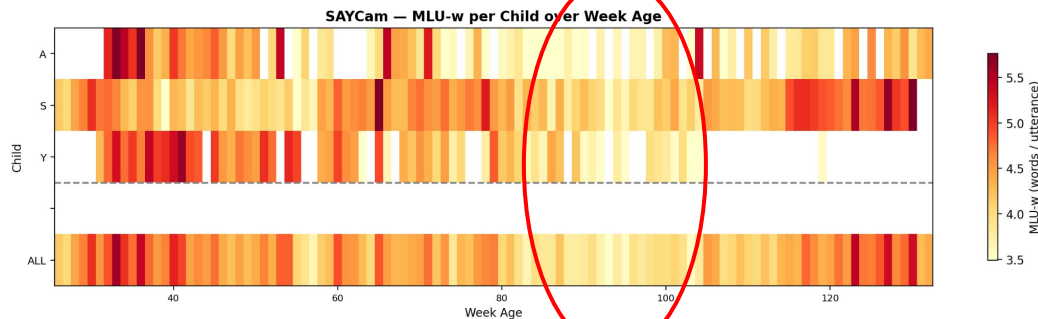
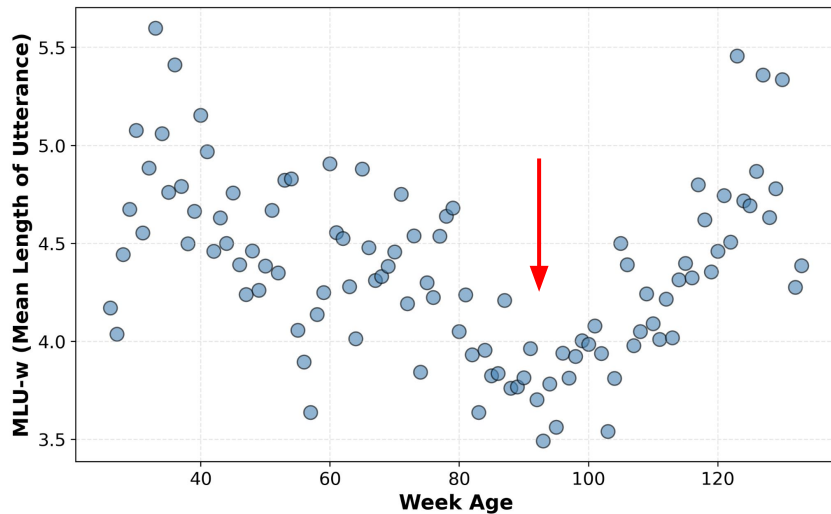


SAYCam dataset - statistical analysis

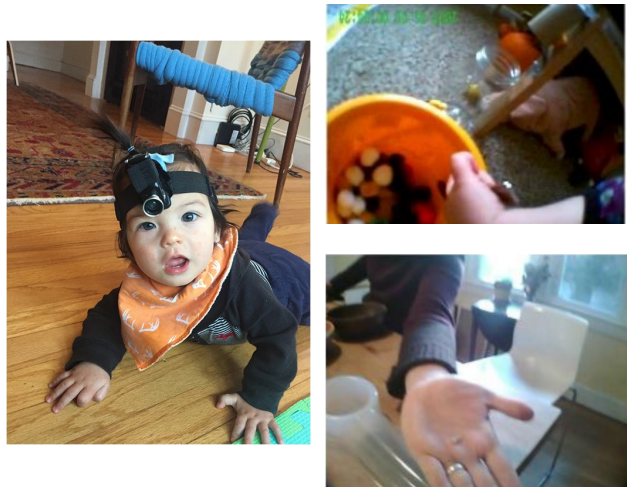
What do babies hear? Utterance analysis

- Transcription: Azure Speech Recognition
- MLU (mean length of utterance): 4.37 words overall

SAYCAM — MLU-w (Mean Length of Utterance) vs Week Age



From infant experience to model-ready data

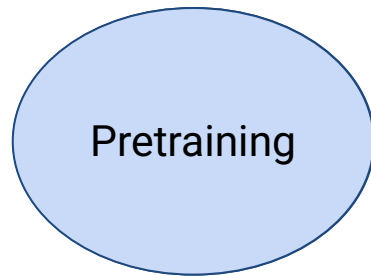
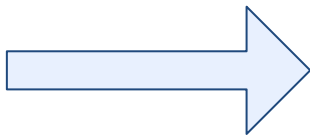


Recorded experience

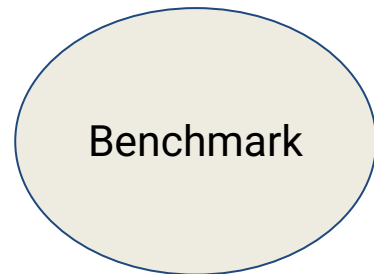
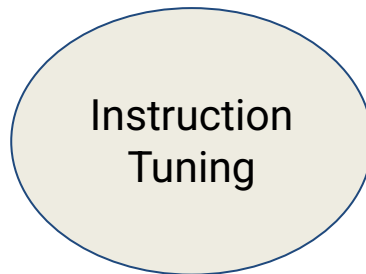
3 infants

478 hours

~6–32 months



Model-ready data



Pretraining data - two signals from the recording

Visual stream



Speech → text

Azure Speech Recognition

Utterance text

Start / end boundaries

Transcription confidence

			
Time: 15:22 Utterance: It's really good with yogurt.	Time: 17:03 Utterance: Want to try a little piece of this?	Time: 18:46 Utterance: How about another blueberry?	Time: 18:59 Utterance: Oh that's a very sweet one.

Pretraining data - split source clips

Split source clips first; derived training examples inherit the source split.

Raw SAYCam clips

Approximate source ratio 3:1:1



Pretraining source

Validation

Test

All clips containing speech

Remaining clips split equally

Pretraining + instruction tuning

**Task
validation**

**Task
testing**

Pretraining data - filter for usable alignment

Filter paired pretraining examples using threshold control

1 Segment speech

One transcript-defined clip per utterance

Duration ≥ 0.5 s

ASR confidence ≥ 0.3

2 Align video + text

X-CLIP similarity

> 0.1

Pad 1s on each side

3 Align frame + text

Sample clips at 1 FPS

CLIP similarity

> 0.2

Pretraining data - three formats

Video-utterance

P1: Video-Utterance Pair



Let's read the first book of sushi
by Amy Wilson Sanger



Image sequence + one utterance

Image-utterance

P2: Image-Utterance Pair



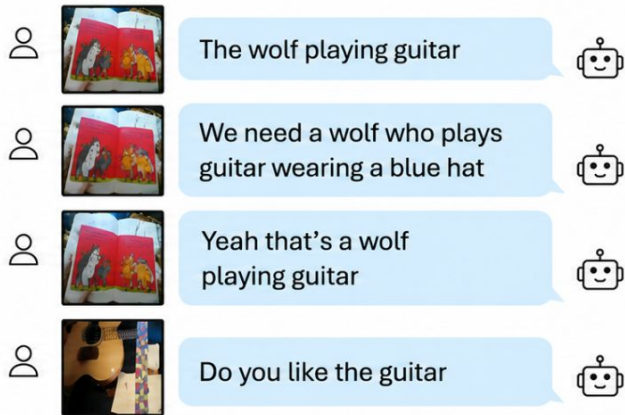
That's the baby dressed as a grape



One frame + one utterance

Interleaved sequence

P3: Multi-turn Image-Utterance



4-8 image-utterance pairs

The resulting pretraining data

Unimodal pools

No alignment filtering

~1.1M

image-only samples

283K

utterance-only
samples

Aligned multimodal formats

768K

image–utterance pairs

181K

video–utterance pairs

63K

interleaved sequences

Developmentally grounded vocabulary



MAB-CDI

MacArthur-Bates
Communicative Development Inventories

Target words or closely aligned concepts

Specialized computer vision tools

Grounding DINO · YOLOE · ByteTrack · ...

Labels, bounding boxes, and temporal events for task construction

Instruction data - labels and boxes become vocabulary questions

Picture Vocabulary



A



B



C



D

Find the stroller.
Reply A, B, C or D.

Target: A

Looking While Listening



A

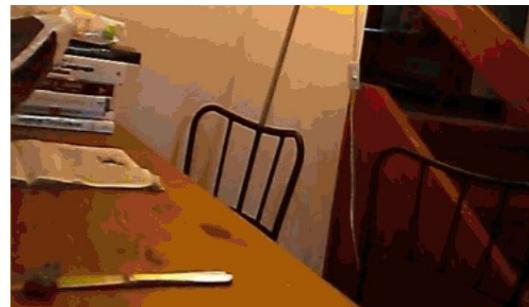


B

Select the chair:
A or B.

Target: B · test-only

Localization



In which quadrant
is the knife?

Target: bottom left

Instruction data - Crops become fine-grained visual matching tasks

Left/Right

Match orientation—not just identity.



Query



A



B



C

Spatial Details

Find the matching occurrence in context.



Query



A



B



C

Which choice matches the query? Reply A, B or C.

Target: B

Target: B

Instruction data - Memory data is an episode—not an isolated pair

Which image is new? Reply A or B. Preserve earlier turns as context.

Learning phase · 5 turns

Turn 1



Turn 2



A



B

New: A

15 distinct images

29 image presentations

Test phase · 10 turns

One learned + one new stimulus



A



B

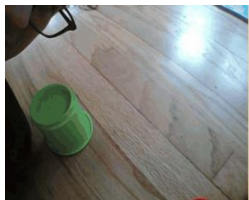
New: B

15 turns

Instruction data - Construct quantities from scenes or object crops

Who Has More

Naturalistic



A



B

Which scene has more cups?

Target: B

Who Has More

Synthetic



A



B

Which image has more balls?

Target: A

Object Counting

1–12 items

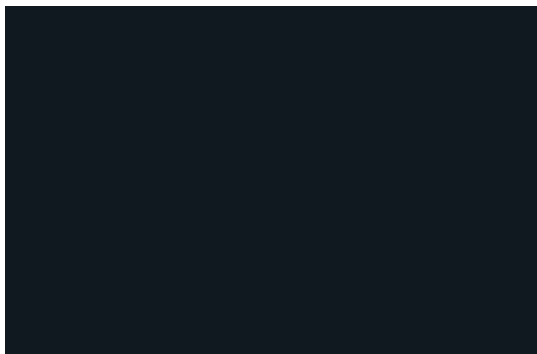


How many balls are shown?

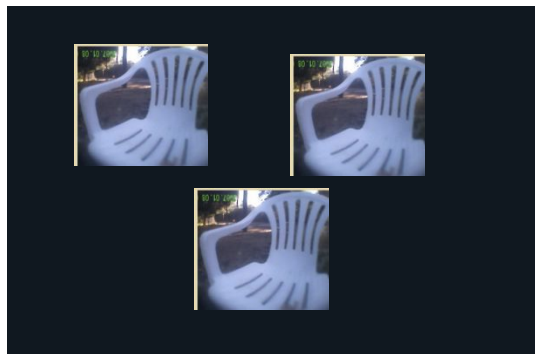
Target: 7

Instruction data - Subitizing uses a brief-stimulus input sequence

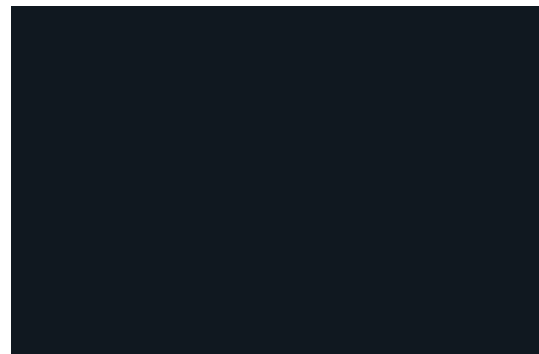
Held out from SFT



Blank frame



Stimulus: 1–4 items



Blank frame

How many chairs were shown?

Target: 3

BabyVLM-v2: Modeling and Algorithm

Toward **Developmentally Plausible** Vision Foundation Models



BabyVLM-V2

Shengao Wang

wsashawn@bu.edu

Presented by

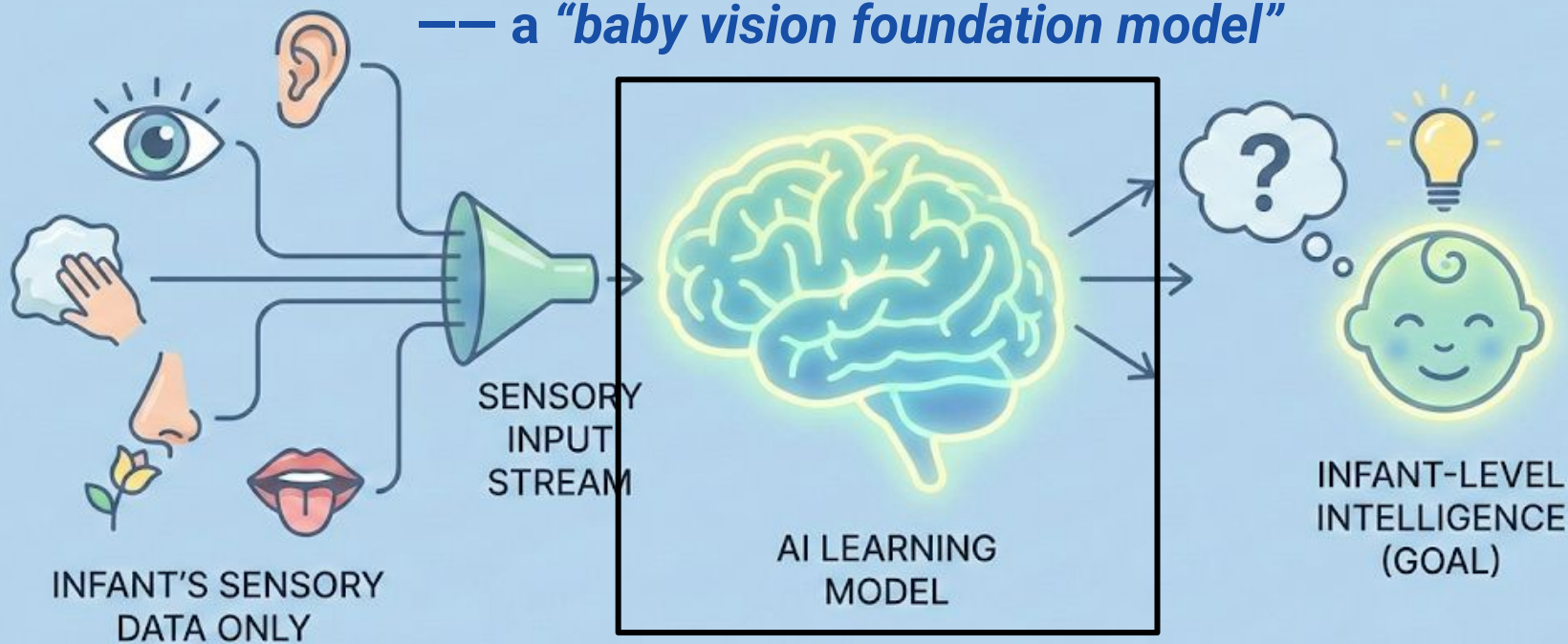
Max Whitton: maxwh@bu.edu



Can we achieve infant-level intelligence using infants' sensory data?

How to model “infant-level intelligence”?

— a “baby vision foundation model”



What is a vision foundation model?

Different model families in different use cases

Generative VLM

LLaVA / Qwen-VL



“What is happening?”



“They are high-fiving.”

Image generation

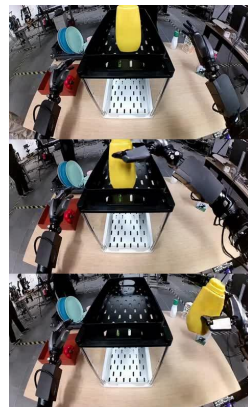
Stable Diffusion

“An astronaut playing piano in space”



World models

Cosmos-Predict2



Earlier



Later

Observation + condition
→ possible future video

What is a vision foundation model?

Different model families in different use cases

Image-text alignment

CLIP



"A dog"



Nearby features in a
shared representation

Match images and text

Visual representations

DINO

Image

Attention map

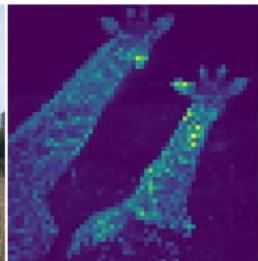


Image → visual features
Useful across visual tasks

3D reconstruction

VGGT

32 Views



0.51 s



Scene views → 3D geometry
Cameras, depth, points

Our focus: generative vision-language models

Image + instruction



“Describe this image.”



**Vision-language
model (VLM)**

e.g., LLaVA or Qwen-VL



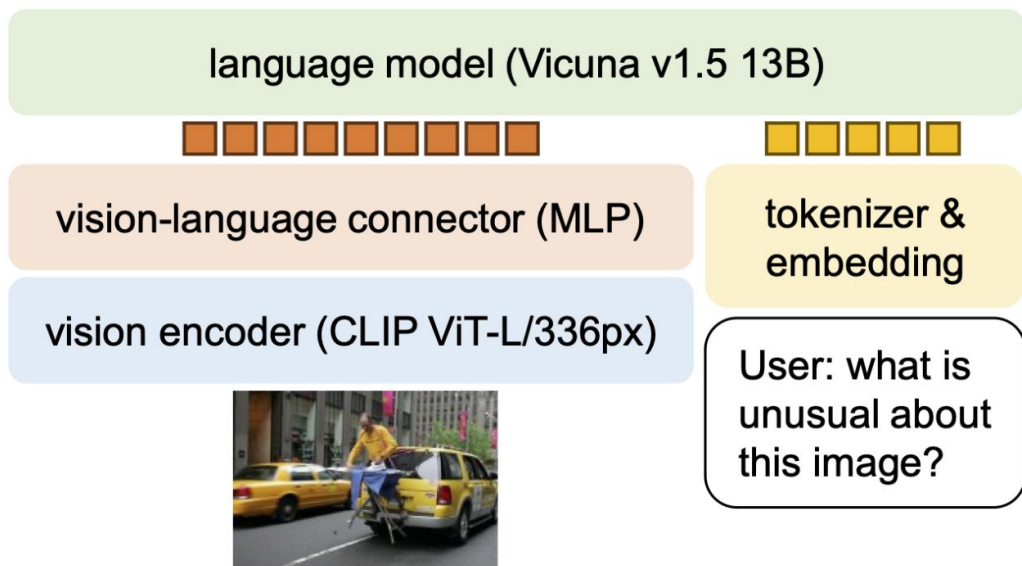
Text response

“A dog and a woman
are giving each other
a high-five on a beach.”

What is a generative vision foundation model?

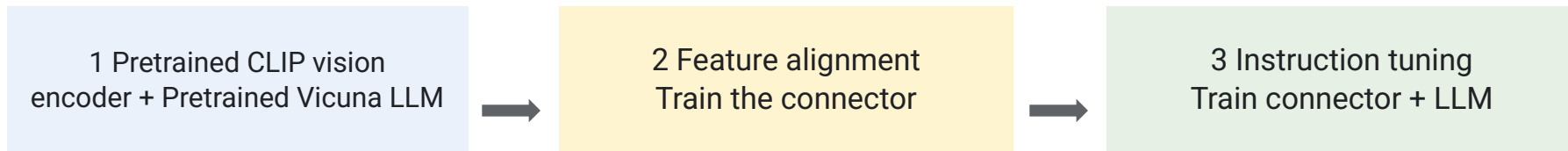
A generative vision foundation model usually comes with three components:

Vision backbone, language backbone, modality connector



How are vision-language models trained?

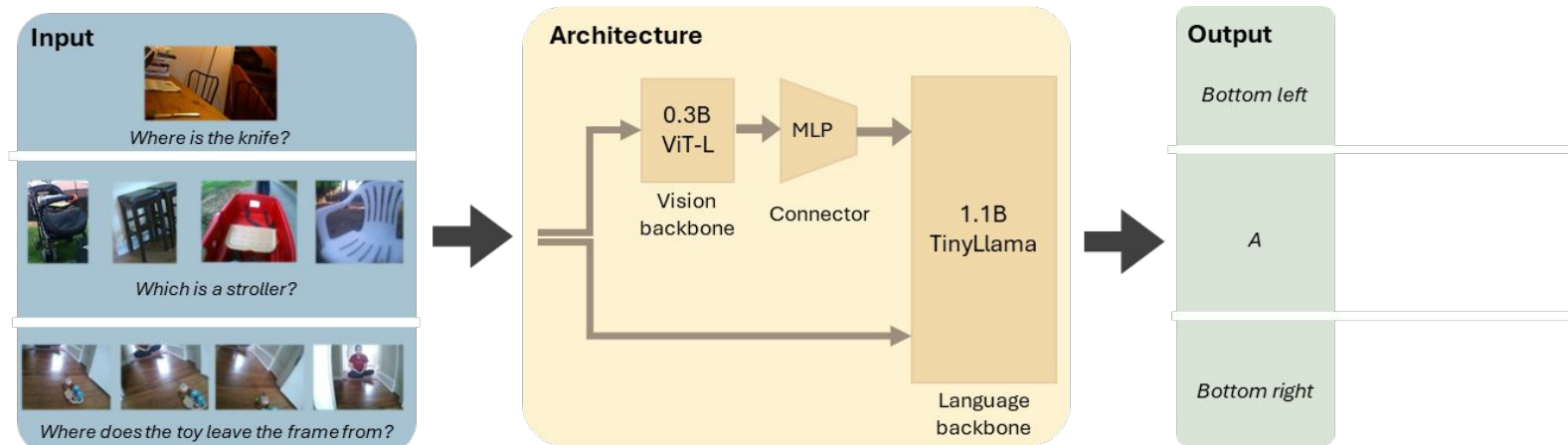
LLaVA: a concrete example



LLaVA-RLHF adds preference training.

Baseline model – Baby model

- **A compact, versatile Vision Language Model (VLM)**
 - Versatile input format: Single image, multiple images, video, multi-turn
 - Follow natural language instruction
- Trained from scratch on our pretrain dataset
- Following LLaVA's architecture & training recipe



Baseline model – Baby model

Input



Where is the knife?

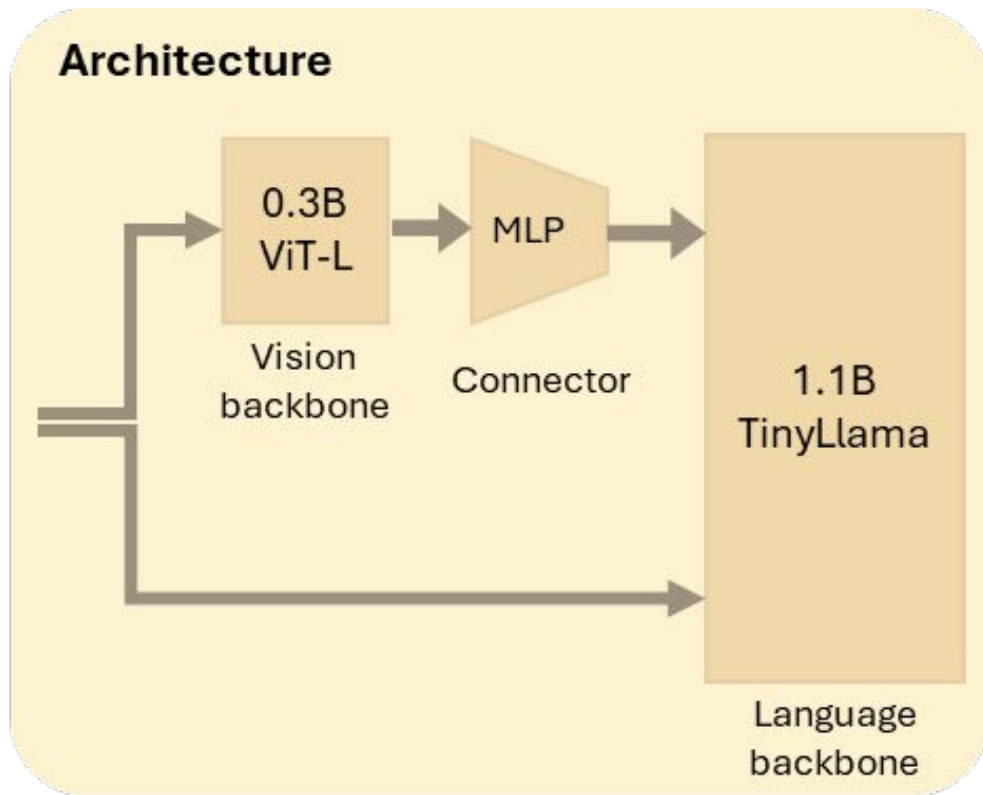


Which is a stroller?



Where does the toy leave the frame from?

Baseline model – Baby model



**Approx. parameters
(one total per model)**

Baby model
≈1.4B total

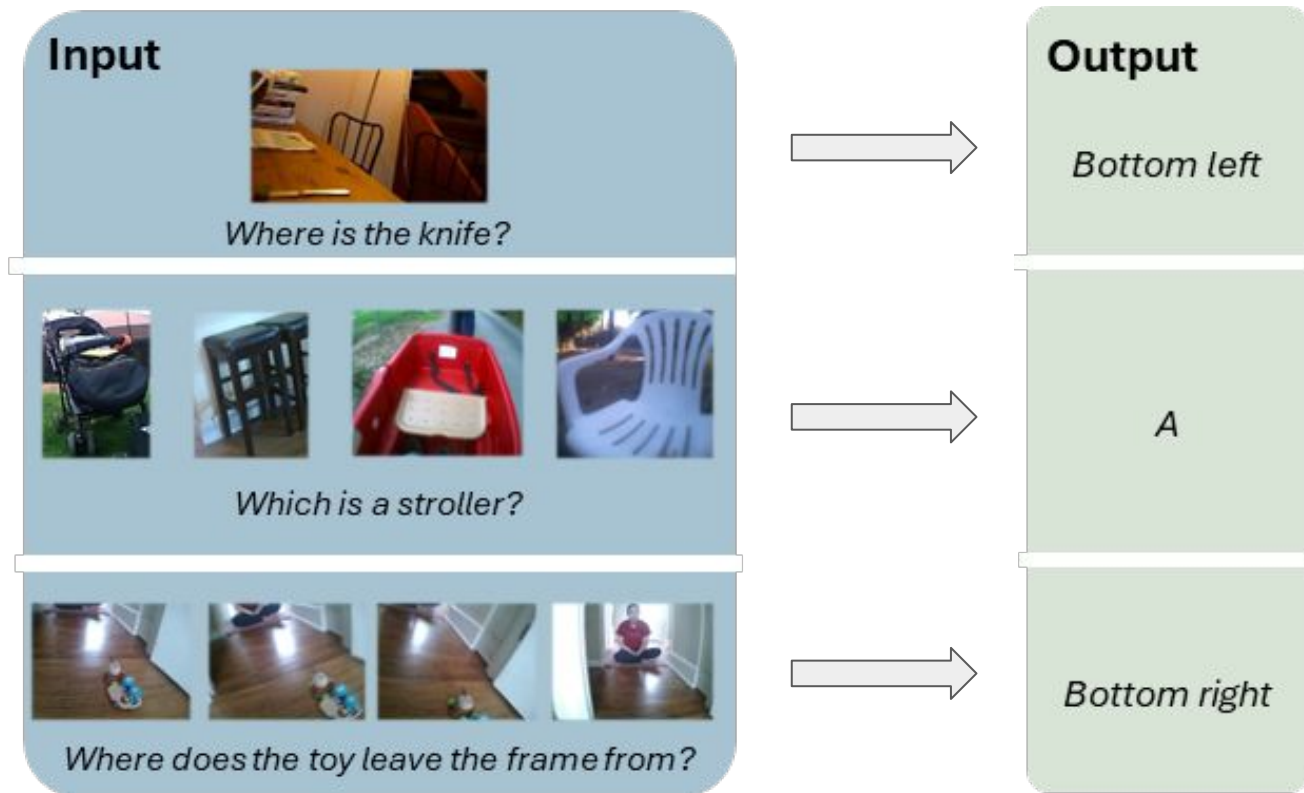
LLaVA-OneVision-7B
≈7B total

Qwen2.5-VL-7B
≈7B total

B = billion parameters. Approximate totals.

Vision backbone recipe: Dosovitskiy, Alexey, et al. "An image is worth 16x16 words: Transformers for image recognition at scale." arXiv preprint arXiv:2010.11929 (2020).
Language backbone recipe: Zhang, Peiyuan, et al. "Tinyllama: An open-source small language model." arXiv preprint arXiv:2401.02385 (2024).

Baseline model – Baby model



Training steps for the baby model

Stage 0 - Language

Unimodal pretrain

Dataset:

- Language only (283k)

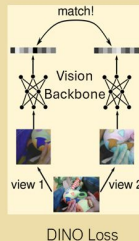


Stage 0 - Vision

Unimodal pretrain

Dataset:

- Vision only (1085k)



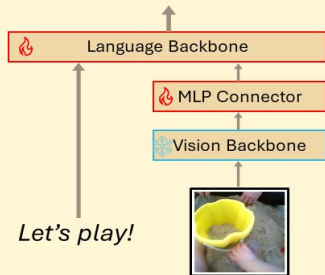
Stage 2

Joint pretrain

Dataset:

- Image-utterance (768k)
- Video-utterance (181k)
- Multi-turn (63k)

Let's play! can you put sand in your cup? and sand in the bucket?



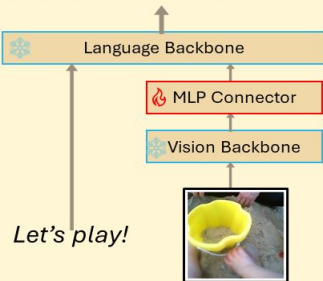
Stage 1

Feature alignment

Dataset:

- Image-utterance (768k)

Let's play! can you put sand in your cup? and sand in the bucket?



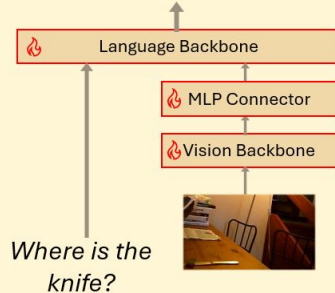
Stage 3

Instruction finetune

Dataset:

- Instruction set (113k)

Bottom left.



Training steps for the baby model

Stage 0 - Language

Unimodal pretrain

Dataset:

- Language only (283k)

*Let's play! can you put sand
in your cup? and sand in the
bucket?*

Language Backbone

Let's play!

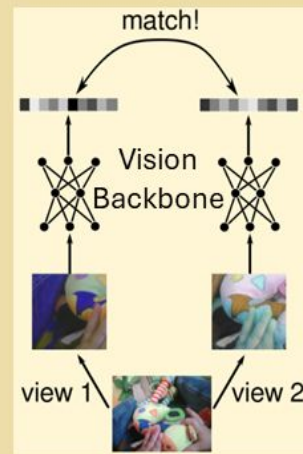
Autoregressive Loss

Stage 0 - Vision

Unimodal pretrain

Dataset:

- Vision only (1085k)



DINO Loss

Training steps for the baby model

Stage 0 - Language

Unimodal pretrain

Dataset:

- Language only (283k)

*Let's play! can you put sand
in your cup? and sand in the
bucket?*

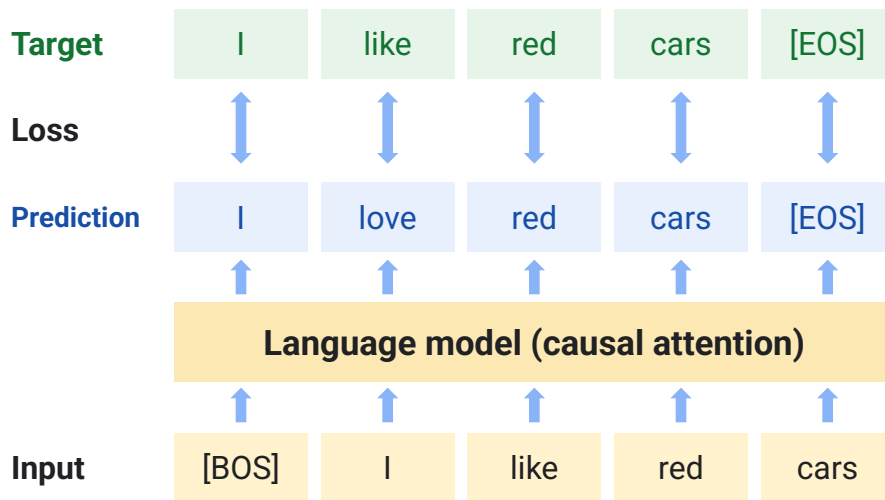
Language Backbone

Let's play!

Autoregressive Loss

Learn to predict the next token

Ground truth: input shifted one position left



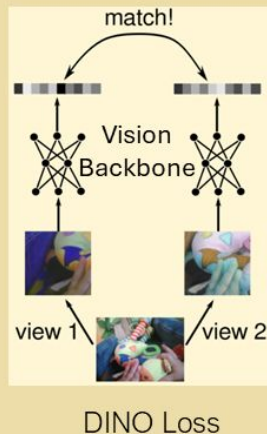
Training steps for the baby model

Stage 0 - Vision

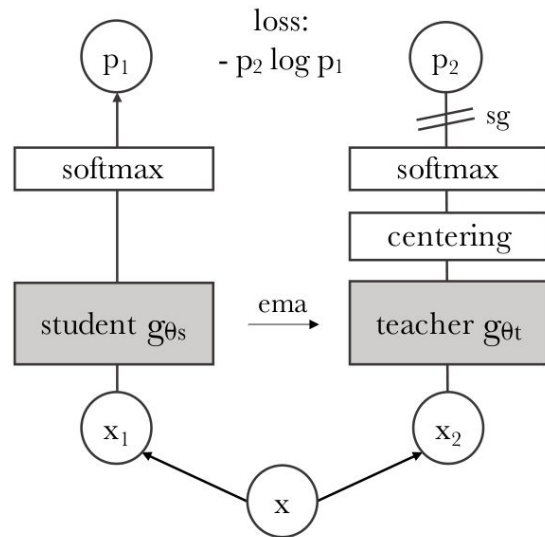
Unimodal pretrain

Dataset:

- Vision only (1085k)



DINO: match two views of one image



Loss compares two output distributions

Train the student to match the teacher.
The targets come from another view—not human labels.

Training steps for the baby model

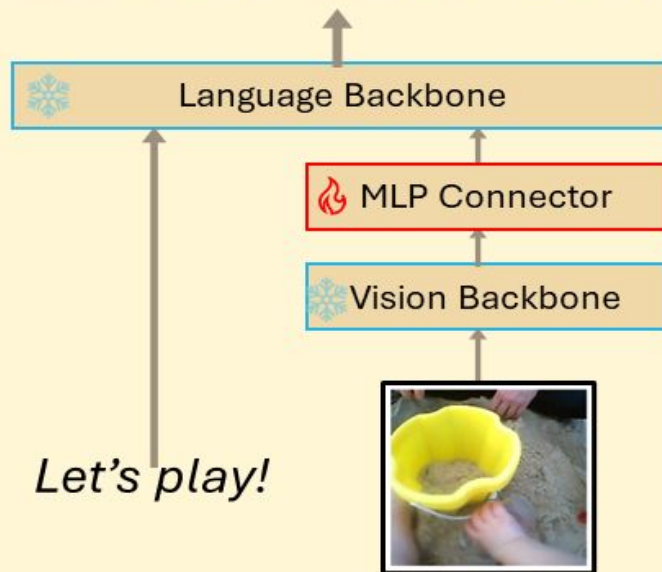
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Feature alignment

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Training steps for the baby model

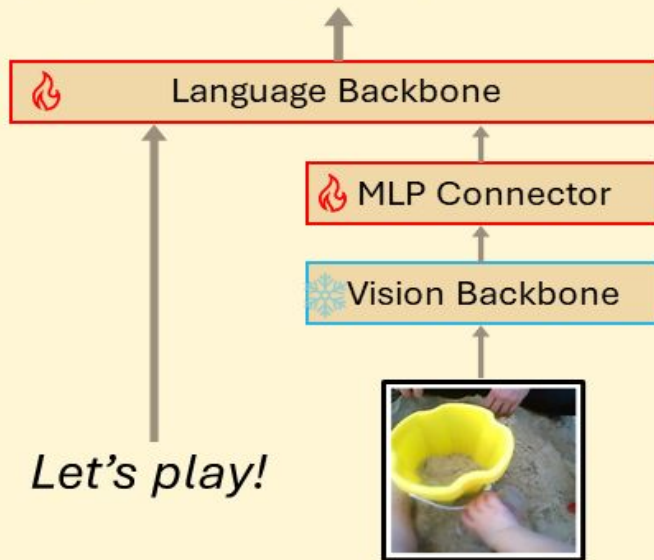
Stage 2

Joint pretrain

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- Image-utterance (768k)
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Let's play! can you put sand in your cup? and sand in the bucket?



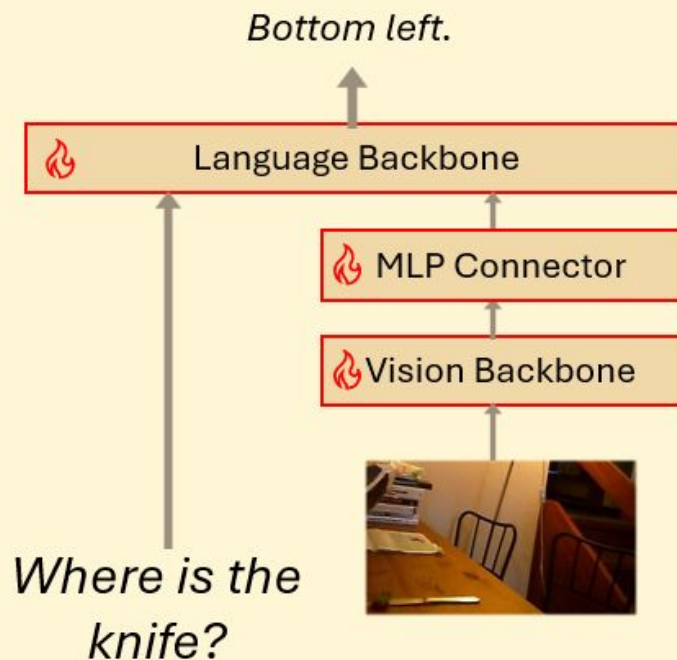
Training steps for the baby model

Stage 3

Instruction finetune

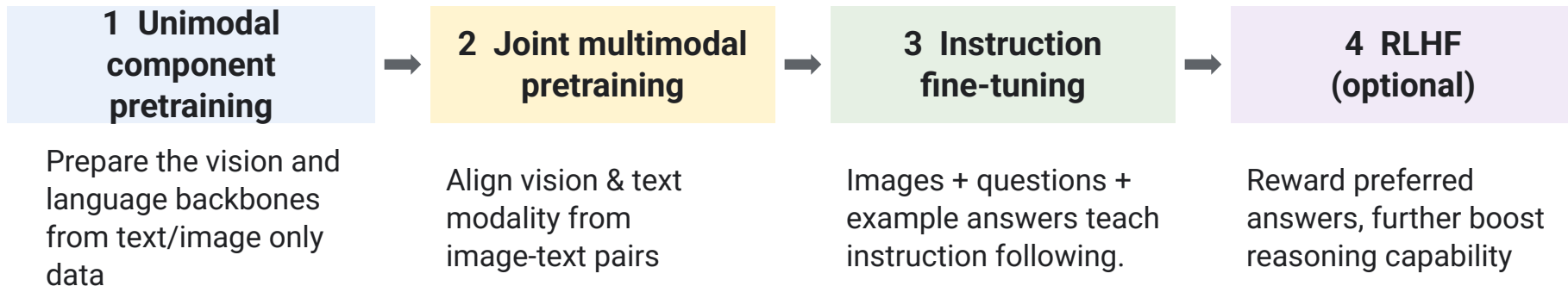
Dataset:

- Instruction set (113k)



How are vision-language models trained?

VLMs first learn initial data pattern from broad data, then learn to follow instructions.



BabyVLM-v2: Developmentally Plausible Benchmarks



BabyVLM-V2

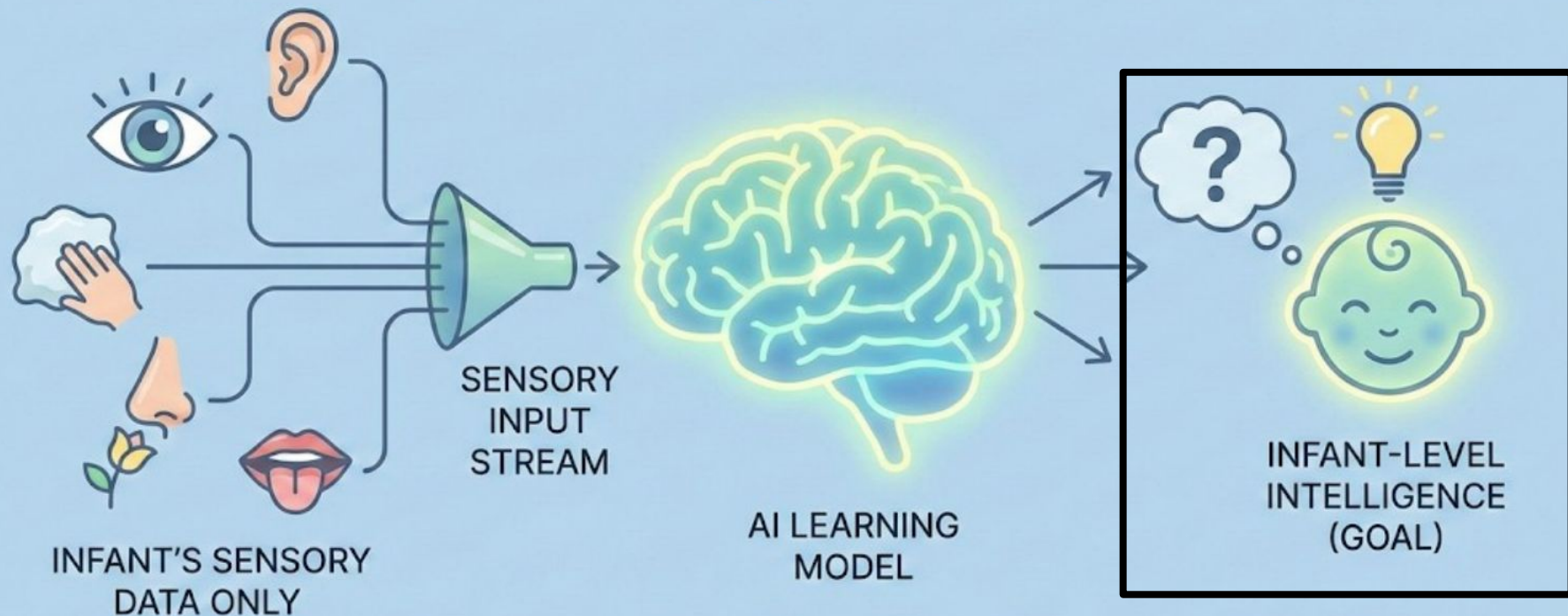
Max Whitton

maxwh@bu.edu

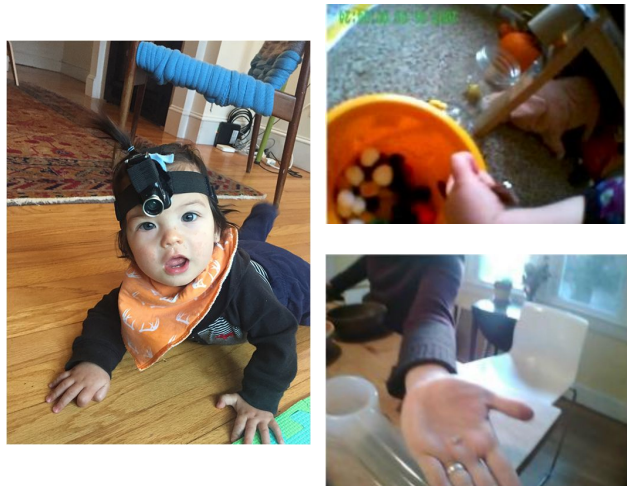


Can we achieve infant-level intelligence using infants' sensory data?

How to assess “infant-level intelligence”?



From infant experience to model-ready data

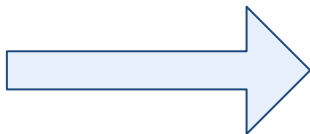


Recorded experience

3 infants

478 hours

~6–32 months



Pretraining

Model-ready data

Instruction
Tuning

Benchmark



Can we achieve infant-level intelligence using infants' sensory data?

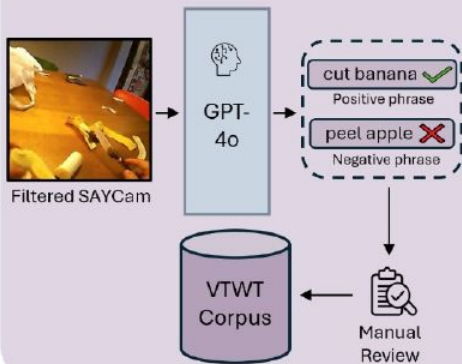
Developmentally **inspired** benchmarks [\[ICCV 2025\]](#) : *Intuitive, heuristic*

Labeled-S

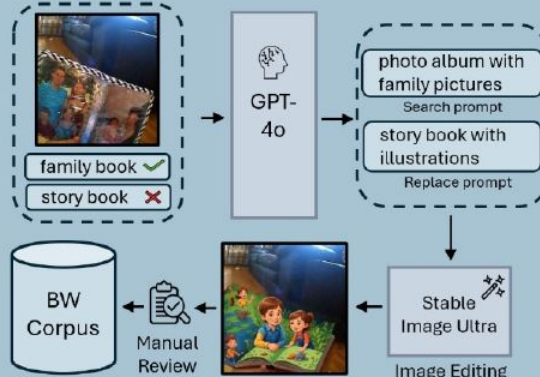


puzzle

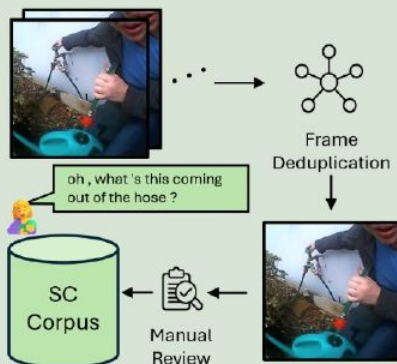
Visual Two-Word Test



Baby Winoground



SAYCam Caption





Can we achieve **infant-level intelligence** using infants' sensory data?

Developmentally **inspired** benchmarks [ICCV 2025] : *Intuitive, heuristic*

Developmentally **aligned** benchmarks

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Can we achieve **infant-level intelligence** using infants' sensory data?

Developmentally **inspired** benchmarks [ICCV 2025] : *Intuitive, heuristic*

Developmentally **aligned** benchmarks

JAN 2025

How to assess
"infant-level
intelligence"?

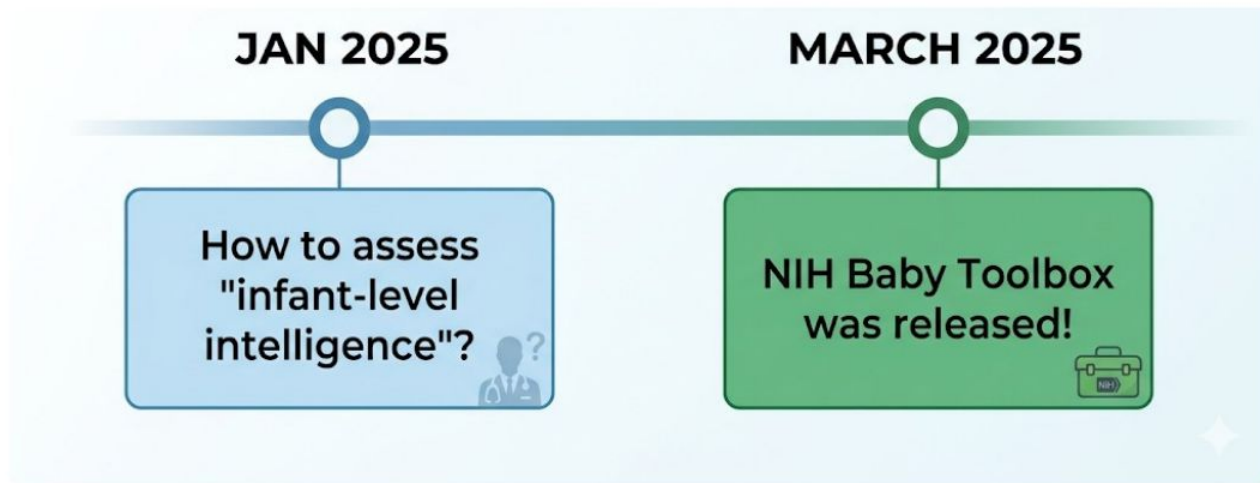




Can we achieve **infant-level intelligence** using infants' sensory data?

Developmentally **inspired** benchmarks [ICCV 2025] : *Intuitive, heuristic*

Developmentally **aligned** benchmarks





Who Is it For?

- Infants & Toddlers
- 1 - 42 months
- Wide Range of Abilities



What Does it Do?

- Measures Cognition, Language, Motor Skills
- Standardized & Normed
- Adaptive Testing



Key Features

- Touchscreen-Tasks/Games
- Short Administration Time
- Provides Standard Scores



Prenatal/Early Infancy:
Baseline Assessment



Longitudinal Monitoring:
Developmental Progress

What is the NIH Baby Toolbox?



Developmental assessment through tablet games



1–42

months of age

31

tests

Cognition
Motor
Social-emotional

Tests matched to developmental age

Who uses the NIH Baby Toolbox?



Educators

School readiness



Clinicians

Developmental
assessment



Research

Developmental
intelligence

Visual delayed response



Does the baby understand
object permanence?

1.5x speed

Can they track moving objects?

Visual delayed response extended, domain-aligned

book



crib



pillow



book



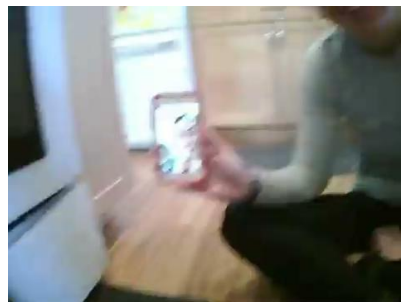
doll stuffed toy



book



phone



doll stuffed toy



DevCV Toolbox



Visual Delayed Response

"Where does the toy leave the frame from? Answer with top or bottom."

Answer: bottom



Memory

Round 1



Rounds 2-15: "Which is the new image? Answer with A or B."

Answer: A



Who Has More

"Which has more cups? Answer with A or B."

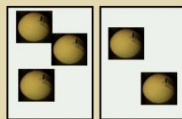
Answer: B



Who Has More

"Which has more balls? Answer A or B."

Answer: A



Subitizing

"How many chairs do you see? Answer with a number 1-12."

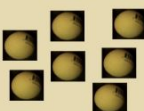
Answer: 3



Object Counting

"How many balls do you see? Answer with a number 1-12."

Answer: 7



NIH Baby Toolbox



Visual Delayed Response

"Look at that."
"Where is it hiding?"

Answer: *touch*



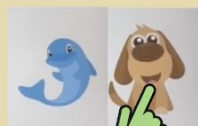
Memory

"The animal is hungry. Touch the animal!"



"Now you see two animals. One of them we already fed. Let's touch the new animal!"

Answer: *touch*



Who Has More?

"Who has more eggs?"

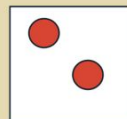
Answer: *touch*



Subitizing



1 second



"How many did you see?"

Answer: 2

Object Counting

"Count the blocks."

Answer: 4



DevCV Toolbox



Picture Vocabulary

"Which is a stroller?
Answer with A, B, C, or D."

Answer: A



Looking While Listening

"Which is a chair?
Answer A or B."

Answer: B



Localization

"Where is the knife?
Answer with top left,
top right, bottom left,
or bottom right."



bottom left

Left / Right



"Look at this cat. Which is exactly
like it? Answer with A, B, or C."

Answer: B



Spatial Details



"Look at this hand. Which
has a hand exactly like it?
Answer with A, B, or C."

Answer: B



NIH Baby Toolbox



Picture Vocabulary

"Touch the picture of chair."

Answer: *touch*



Looking While Listening

"Touch the picture of
of table."

Answer: *touch*



Mullen RL #19

"Look at these.
Where is the cat?"

Answer: *point*



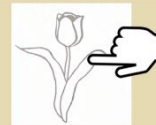
Mullen VR # 29

"Look at this boy. Find another
boy that is just like this one."

Answer: *point*



Mullen VR # 25



"Look at this flower. Look for one
just like it. Look for it here."

Answer: *point*



Memory



Can the baby remember objects, without them being named?

Memory

"The animal
is hungry.
Touch the
animal!"



"Now you see two animals.
One of them we already fed.
Let's touch the new animal!"

Answer: *touch*



Memory



Memory

Round 1



Rounds 2-15: "Which is the new image? Answer with A or B."

Answer: A



Who Has More?



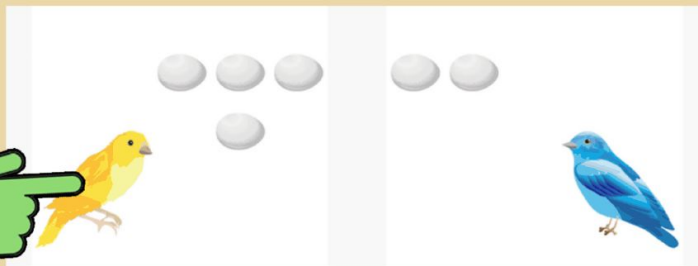
Can the baby compare two quantities of the same object?

Being able to count is not required!

Who Has More?

"Who has more eggs?"

Answer: *touch*



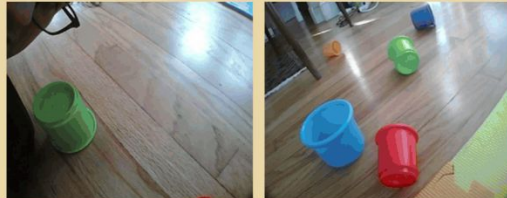
Who Has More?



Who Has More

"Which has more cups? Answer with A or B."

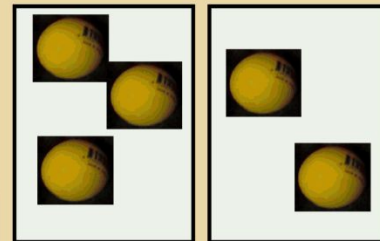
Answer: B



Who Has More

"Which has more balls? Answer A or B."

Answer: A



Object Counting



Can the baby count to 12?

The toolbox offers partial credit for counting consecutively up to a point, which we plan to implement for future work.

Object Counting

"Count the
blocks."

Answer: 4



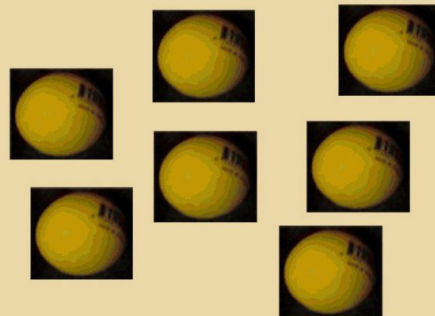
Object Counting



Object Counting

"How many balls do you see? Answer with a number 1-12."

Answer: 7



Picture Vocabulary



Can the baby recognize objects by name?

Extra challenge: the distractors are semantically related!

Picture Vocabulary

"Touch the picture of chair."

Answer: *touch*



Picture Vocabulary



Picture Vocabulary

“Which is a stroller?
Answer with A, B, C, or D.”

Answer: A



Localization



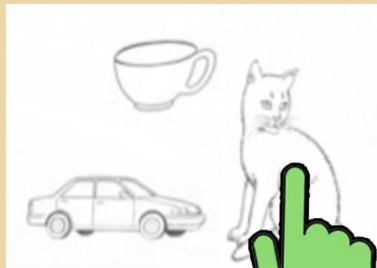
Can the baby point at a named object?

Extra challenge: distractor objects

Mullen RL #19

"Look at these.
Where is the cat?"

Answer: *point*



Localization



Localization

“Where is the knife?
Answer with top left,
top right, bottom left,
or bottom right.”



bottom left

Mullen Visual Reasoning #29

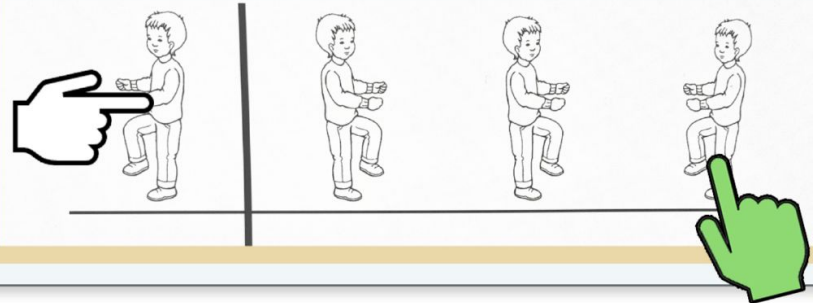


Can the baby match objects by orientation?

Mullen VR # 29

"Look at this boy. Find another boy that is just like this one."

Answer: *point*



Left/Right



Left / Right



"Look at this cat. Which is exactly like it? Answer with A, B, or C."

Answer: B



Mullen Visual Reasoning #25



Can the baby remember the details of a particular instance of an object versus others from its category?

Mullen VR # 25



"Look at this flower. Look for one just like it. Look for it here."

Answer: *point*



Spatial Details



Spatial Details



“Look at this hand. Which
has a hand exactly like it?
Answer with A, B, or C.”

Answer: B





Can we achieve **infant-level intelligence** using infants' sensory data?

Developmentally **aligned** benchmarks

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BabyVLM-v2: Instruction Tuning

Toward **Developmentally Plausible** Vision Foundation Models

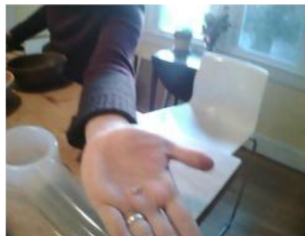


BabyVLM-V2

Wenqi Wang

wqwang@bu.edu

From infant experience to model-ready data

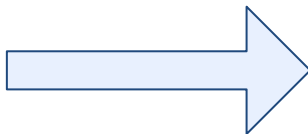


Recorded experience

3 infants

478 hours

~6–32 months



Pretraining

Model-ready data

Instruction
Tuning

Benchmark

What is instruction tuning?

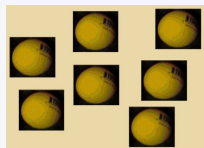
Instruction tuning

Use instructions + target responses to teach instruction following.

SFT: supervised fine-tuning

Update a pretrained model using labeled input–target examples.

In BabyVLM-v2, instruction tuning is done with SFT.



“How many balls?”

Image + instruction

+

Target

7



Update weights

Learn the target response

Training: target provided. Evaluation: answer hidden.

Why do we need instruction tuning?

In Pretraining



“Can you do
the white one?”

Naturalistic language—not explicit task

In benchmarking



How many balls?
Answer with 1–12.

Target: 7

Visual input + instruction + target

Mismatch!

SFT updates a pretrained model using labeled input–target examples.

Here, instruction tuning teaches task instructions and answer formats.

Instruction data - annotations become task supervision

1 Retrieve evidence

Labels, regions, tracks

2 Construct the task

Choices, crops, or sequences

3 Create supervision

Input + instruction → target

Language grounding

Labels + crops + box centers

**Picture Vocabulary · Looking While
Listening · Localization**

Visual matching

Crops + orientation + context

Left/Right · Spatial Details

Temporal / memory

Object events or trial history

Visual Delayed Response · Memory

Quantity

Natural counts or repeated
crops

**Who Has More · Subitizing
Object Counting**

Single image, multiple images, or multiple turns — one text interface.

Instruction data - annotations become task supervision

Example: *picture vocabulary* task construction

Picture Vocabulary

"Which is a stroller?
Answer with A, B, C, or D."

Answer: A



The image shows a task interface for a 'Picture Vocabulary' task. It contains a question: 'Which is a stroller? Answer with A, B, C, or D.' Below the question is a green button with the text 'Answer: A'. To the right of the text are four small images labeled A, B, C, and D. Image A shows a black stroller. Image B shows a wooden chair. Image C shows a red plastic chair. Image D shows a blue plastic chair.



Frames sampled from
SAYCam

We develop data samples constructor for each task in DevCV Toolbox

Instruction data - 10 tasks, 2 held out w./o SFT data

≈113K

Supervised Finetuning

≈13.6K

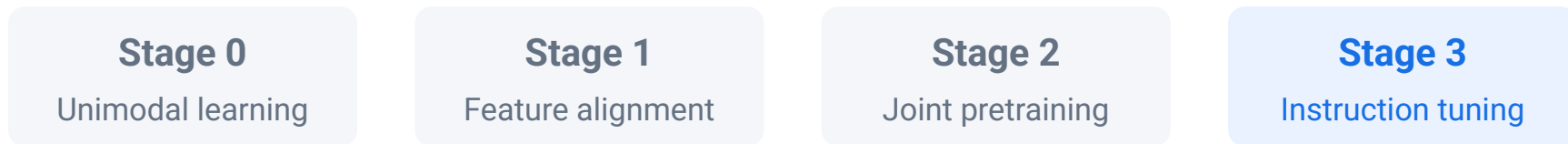
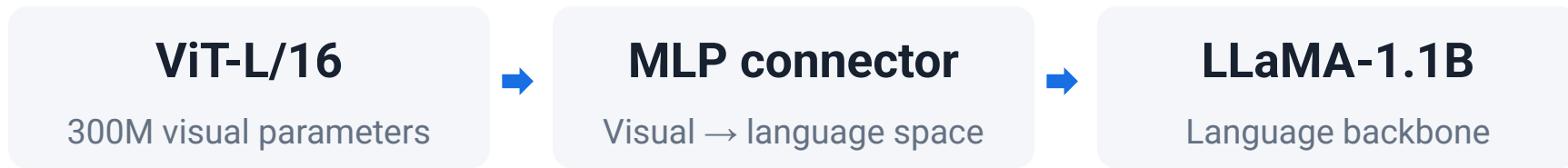
Validation

≈17.8K

Test (DevCV benchmark)

Task	SFT (K)	Validation (K)	Test (K)
Picture Vocabulary	22.4	0.7	1.1
Looking While Listening	0	0	1.2
Localization	12.3	2.1	2.1
Left/Right	12.3	2.2	2.3
Spatial Details	11.8	1.2	1.2
Visual Delayed Response	2.6	0.4	0.5
Memory	12.0	0.3	0.3
Who Has More (synthetic / natural)	11.2 / 12.4	1.8 / 1.9	1.8 / 2.2
Subitizing (synthetic / natural)	0 / 0	0 / 0	1.9 / 0.2
Object Counting	13.7	3.0	3.0

Setup - A compact model, with SFT as the final stage



Stage 3 updates all modules with an autoregressive objective.

5 epochs

Global batch: 128

$1e-4$ / $5e-4$

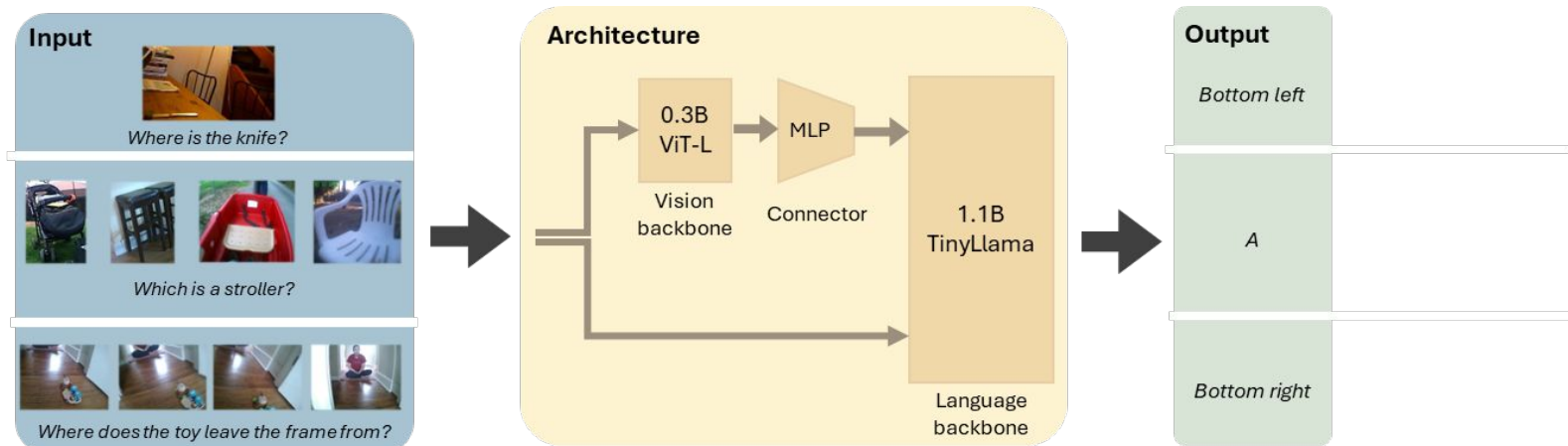
Vision / connector + language

≈24 hours

4 × A6000, 48 GB each

Baseline model – Baby model

- **A compact, versatile Vision Language Model (VLM)**
 - Versatile input format: Single image, multiple images, video, multi-turn
- Trained from scratch on our pretrain dataset
- Following LLaVA's architecture & training recipe



What is instruction tuning?

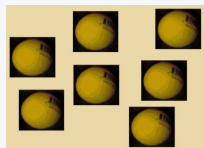
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In BabyVLM-v2, instruction tuning is done with SFT.



“How many balls?”

Image + instruction

+

Target

7



Update weights

Learn the target response

Training: target provided. Evaluation: answer hidden.

Why we need instruction tuning?

What the model hears

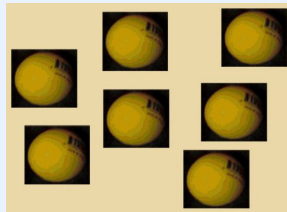


“Can you do
the white one?”

Natural speech—not task answers.



What DevCV asks for



How many balls?
Answer with 1–12.

7

Explicit prompt + target answer

Instruction tuning connects naturalistic representations to task instructions and answer formats.

SFT data - Annotations become task supervision

1 Retrieve evidence

Concepts, boxes, tracks



2 Construct the task

Target + choices / sequence



3 Create supervision

Input + prompt → target

Task	How the annotation determines the answer	Target
Picture Vocabulary	Target concept + position among four choices	A
Localization	Bounding-box center → image quadrant	bottom left
Visual Delayed Response	Human-reviewed direction of disappearance	bottom
Object Counting	Known number of pasted object crops	7
Memory	Previously seen images + new-image identity	A / B

SFT data - use text interface for different tasks

Multiple images



A



B



C



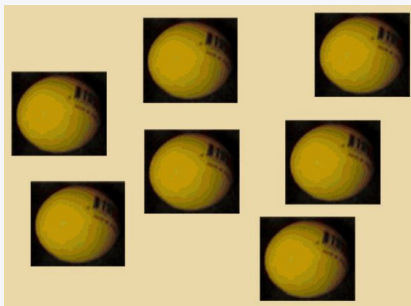
D

Which is a stroller?

Answer A, B, C or D.

Target: A

Single image



How many balls?

Answer with 1–12.

Target: 7

Multiple turns

Previously seen



A



B

Which image is new?

Target: A

BabyVLM-v2: Evaluation & Experiments

Toward **Developmentally Plausible** Vision Foundation Models



BabyVLM-V2

Wenqi Wang

wqwang@bu.edu

Evaluation — In-domain and two domain shifts

Baby model: SAYCam pretraining + mixed instruction tuning

SAYCam(in-domain)



In-domain

Infant-egocentric imagery
Held-out source split
Adapted DevCV tasks

Ego4D(out of domain)



New naturalistic domain

Adult-egocentric imagery
Different environments
Same task construction

Original NIH Toolbox (out of domain)



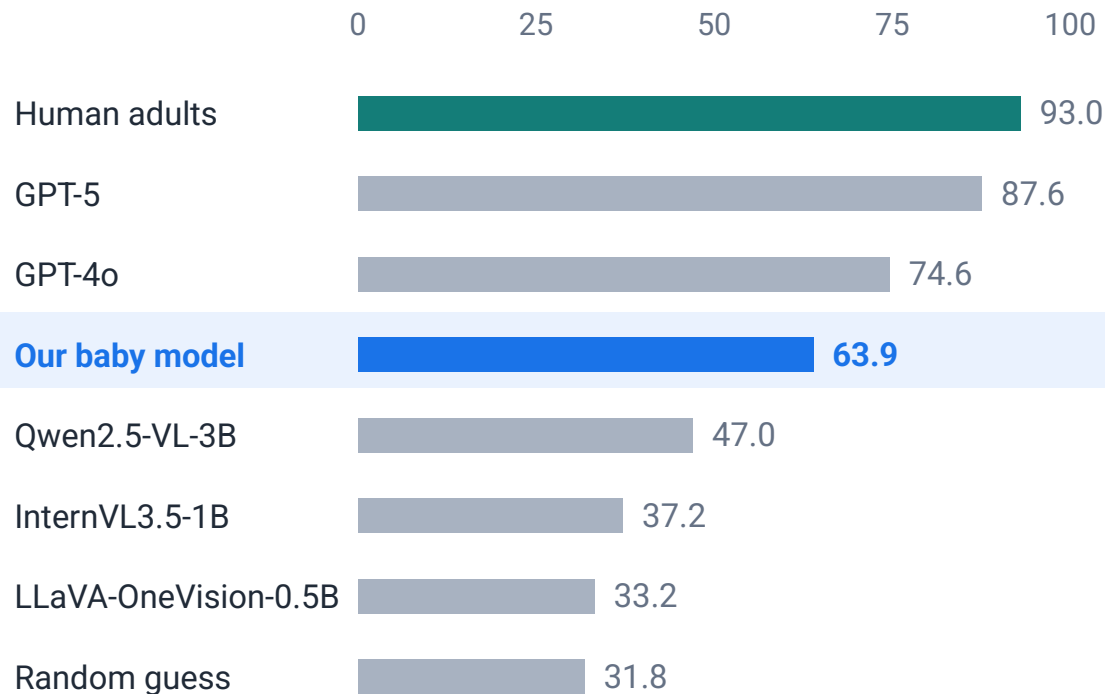
Original stimuli

Cartoon / abstract imagery
Small task-specific sets
Report correct / total

Held out from SFT: Looking While Listening and Subitizing.

Main result — 63.9% on in-domain DevCV

Selected reference models and the adult survey



63.9%

Overall accuracy

Above the compact checkpoints shown.

Not matched for data or adaptation.

Main result — An uneven skill profile

Task accuracy (%) · gap = baby model minus GPT-4o

Task	Baby model	GPT-4o	Gap (pp)
Object Counting	47.3	39.0	+8.3
Left/Right	96.4	89.8	+6.6
Spatial Details	92.8	92.6	+0.2
Who Has More — synthetic	99.7	87.9	+11.8
Picture Vocabulary	32.4	93.7	-61.3
Localization	37.8	81.7	-43.9
Who Has More — naturalistic	60.5	79.3	-18.8

Controlled synthetic comparisons do not guarantee success in natural scenes.

Validation – Instruction data helps other VLMs

A separate validation experiment: one task-specific model per task.

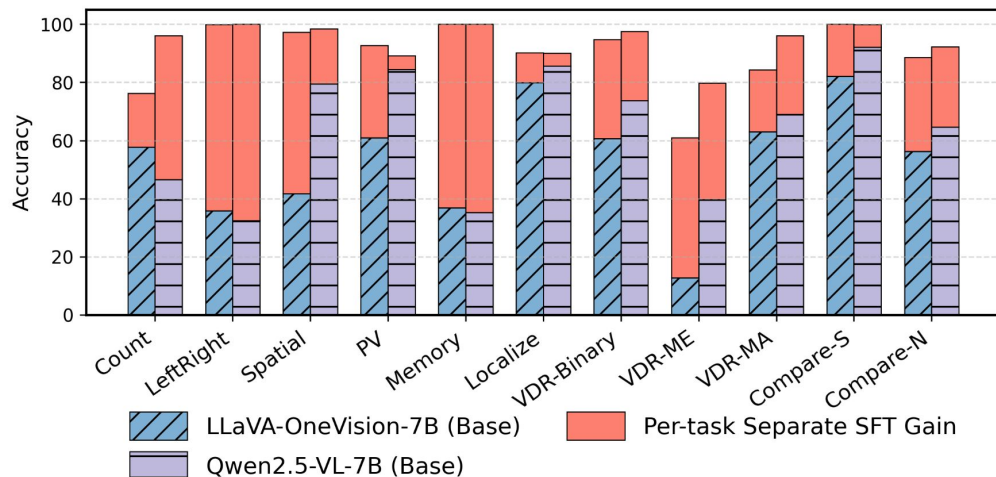
LLaVA-OneVision-7B/
Qwen2.5-VL-7B



Task-specific
LoRA fine-tuning



Gains across
tasks

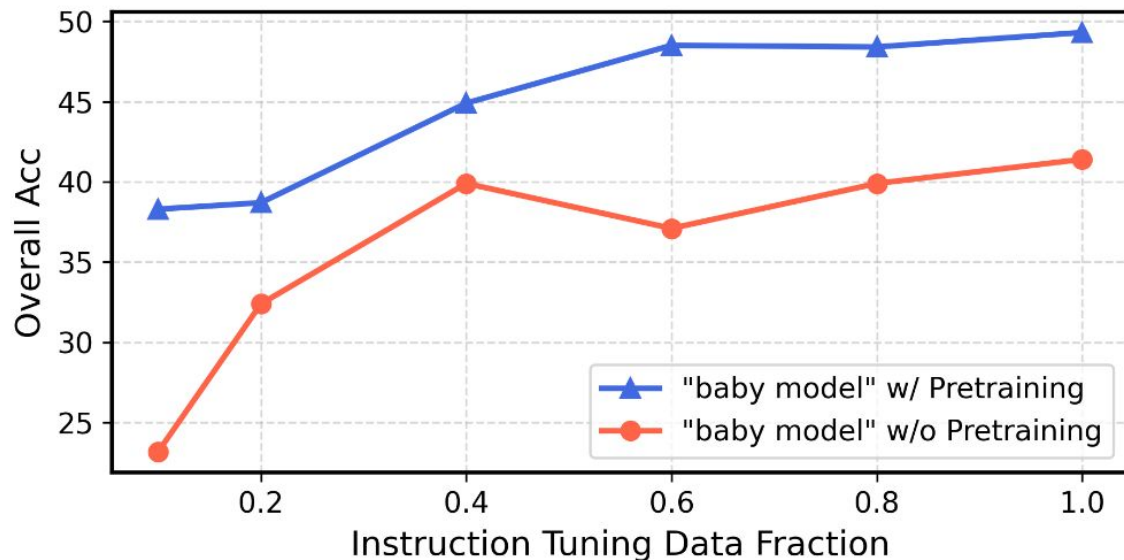


5 epochs · batch 128 ·
learning rate 1e-4

Useful supervision—effectively
guides models toward the
tasks in DevCV Toolbox.

Ablation — Pretraining helps data efficiency

Same architecture · vary the instruction-data fraction · evaluate after one epoch



**Pretraining wins
at every fraction.**

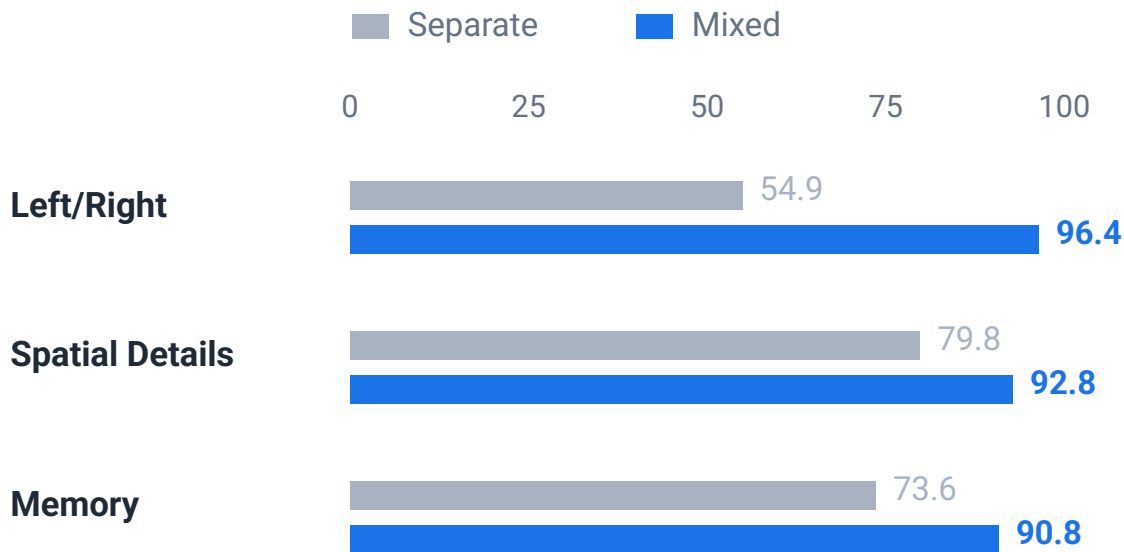
Largest advantage
when instruction
data is limited.

Stages 0–2 + SFT
vs. SFT from scratch

Task supervision complements earlier experience; it does not replace it.

Ablation — Mixed-task SFT improves the overall score

Separate task-specific models vs. one jointly tuned model



Overall accuracy

56.9 → 63.9

+7.0 points

One model,
multiple task formats.

Not every metric improves: naturalistic Who Has More, 62.1 → 60.5.

Ablation — Natural caregiver speech remains competitive

Change the pretraining language; keep the architecture and downstream SFT fixed.

Overall DevCV accuracy (%)

Caregiver utterances  **63.9**

GPT-4o captions  **65.4**

+1.5 points

Only a modest overall gain.



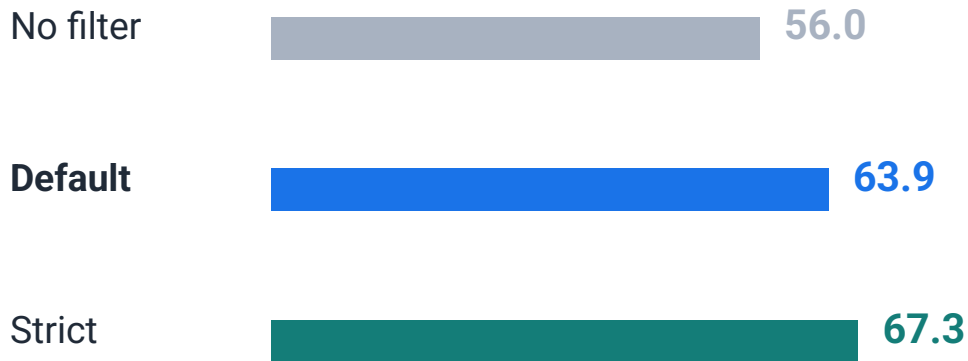
GPT-4o: Look at the cute chalk drawing!

Original: No I drew his coat on so I protected him from the mud.

Ablation — Filtering quality matters

Same model and downstream SFT; different pretraining filters

Overall DevCV accuracy (%)



67.3%

Strict filtering

Less than half
the default
pretraining data.

Better retained pairs—not simply more pretraining pairs.

Limitation — Transfer across visual domains is limited

The same task builders, but different visual experience

SAYCam → Ego4D

Metric	SAYCam	Ego4D	Random
Overall	63.9	48.8	31.8
Spatial Details	92.8	47.1	33.3
Memory	90.8	72.4	25.0

Original NIH Toolbox

Correct / total · small sets

Who Has More	13 / 24
Object Counting	2 / 6
Mullen Visual Reception	3 / 12

Above chance on Ego4D, but the in-domain skill profile does not fully transfer.

Limitation — Related tasks do not yet transfer

Two task families receive no instruction-tuning examples.

Seen during SFT

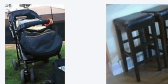
Held out from SFT

Outcome

Picture Vocabulary



Looking While Listening



4 → 2 choices

Near chance

50% random

Object Counting



1 image
1–12 objects



Subitizing

Blank

1–4

Blank

Near chance

25% random

References

- Wang et al. (2026). BabyVLM-V2: Toward Developmentally Grounded Pretraining and Benchmarking of Vision Foundation Models. CVPR.
- Wang et al. (2025). BabyVLM: Data-Efficient Pretraining of VLMs Inspired by Infant Learning. ICCV.
- Vong et al. (2024). Grounded language acquisition through the eyes and ears of a single child. Science, 383(6682), 504–511.
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- Gershon, Novack & Kaat (2024). The NIH Infant and Toddler Toolbox: A new standardized tool for assessing neurodevelopment in children ages 1–42 months. Child Development, 95, 2252–2254.
- Long et al. (2025). The BabyView dataset: High-resolution egocentric videos of infants’ and young children’s everyday experiences. Cognitive Computational Neuroscience Conference 2025
- Grauman et al. (2022). Ego4D: Around the World in 3,000 Hours of Egocentric Video. CVPR.
- Liu et al. (2024). Improved Baselines with Visual Instruction Tuning. CVPR. [LLaVA-1.5]
- Bai et al. (2025). Qwen2.5-VL Technical Report. arXiv:2502.13923.
- Radford et al. (2021). Learning Transferable Visual Models From Natural Language Supervision. ICML. [CLIP]

Q&A

Evaluation & Experiments

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BabyVLM-V2

Evaluation - In-domain and two domain shifts

Baby model: SAYCam pretraining + mixed instruction tuning

SAYCam

In-domain

Infant-egocentric imagery
Held-out source split
Adapted DevCV tasks

Ego4D

New naturalistic domain

Adult-egocentric imagery
Different environments
Same task construction

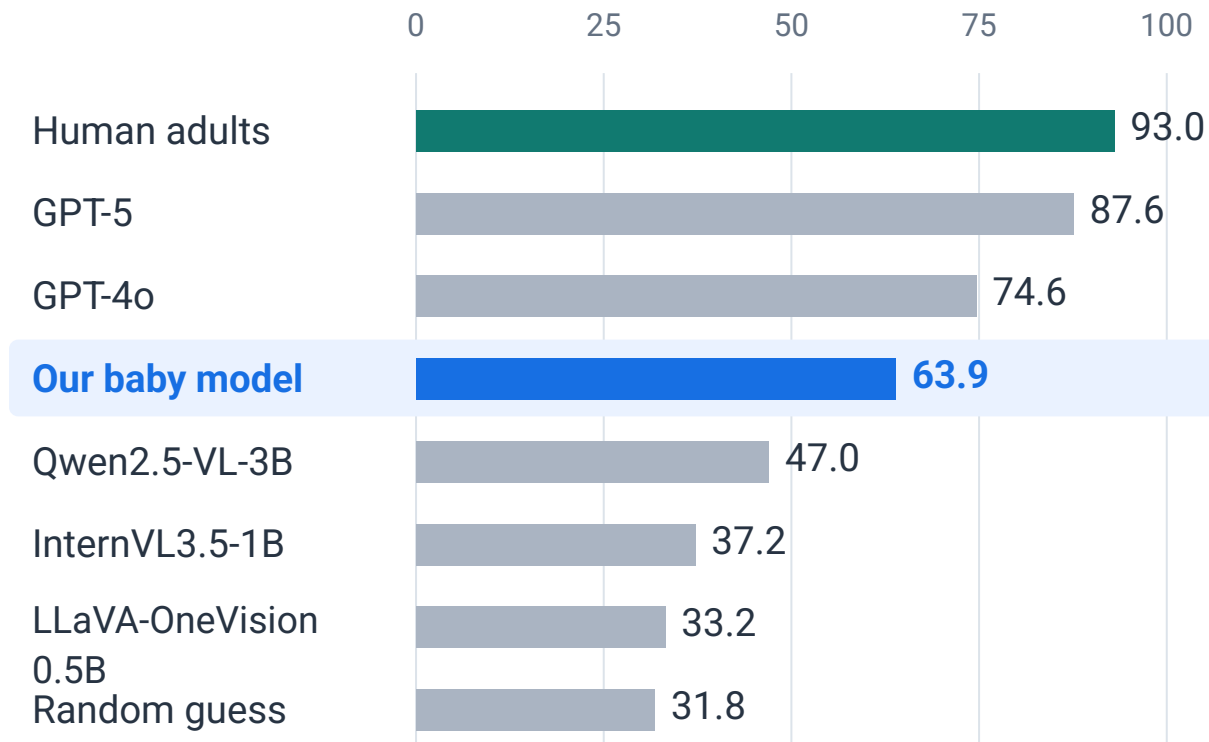
Original NIH Toolbox

Original stimuli

Cartoon stimuli
Small test sets
Correct / total counts

A separate axis: hold out Looking While Listening and Subitizing from SFT.

Main result - 63.9% on in-domain DevCV



63.9%

**Overall
accuracy**

Competitive with
compact public
checkpoints.

Not an equal-data
or equal-SFT
comparison.

Main result - An uneven skill profile

Task	Baby model	GPT-4o	Gap (pp)
Object Counting	47.3	39.0	+8.3
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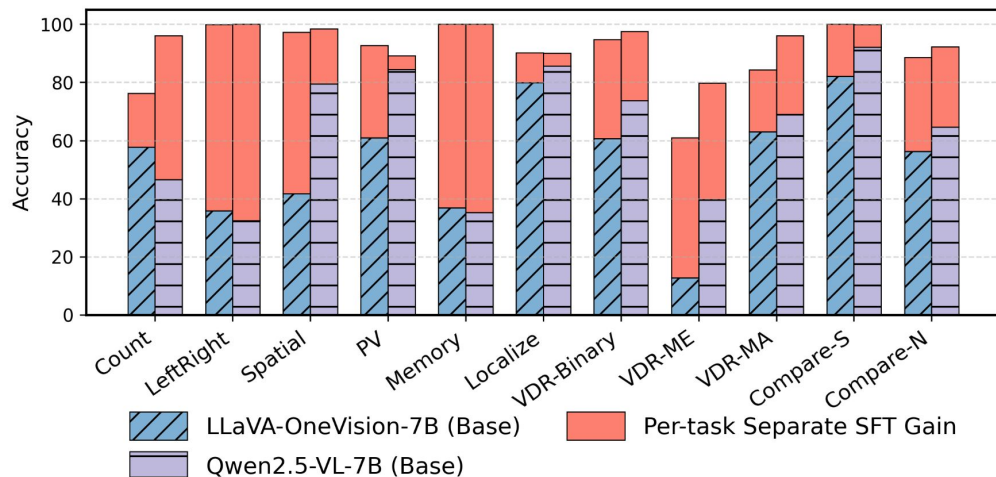
**LLaVA-OneVision-7B/
Qwen2.5-VL-7B**



**Task-specific
LoRA fine-tuning**



**Gains across
tasks**

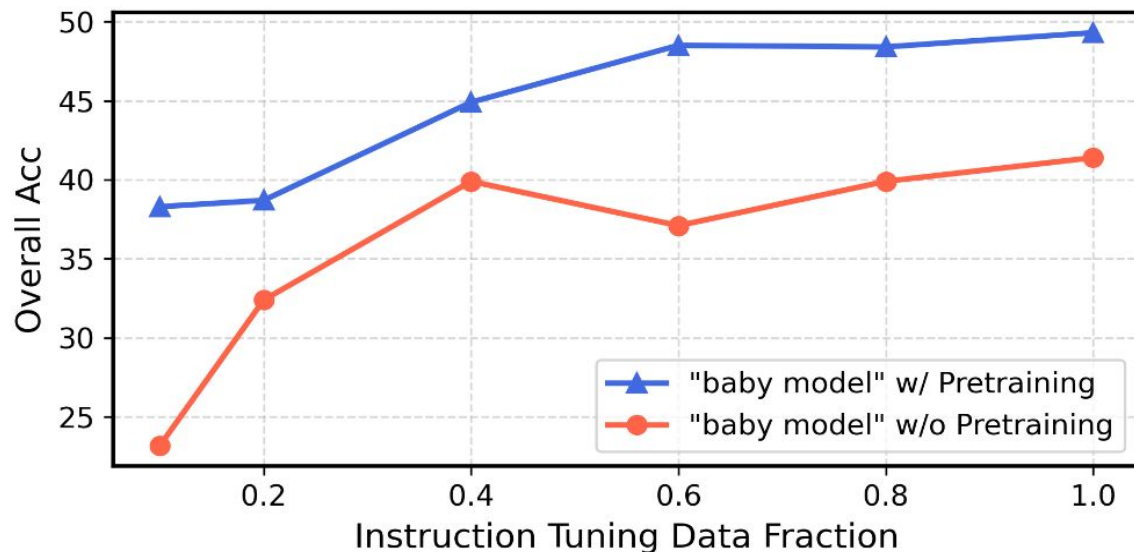


5 epochs · batch 128 ·
learning rate 1e-4

**Useful supervision—effectively
guides models toward the
tasks in DevCV Toolbox.**

Ablation - Pretraining helps with limited SFT data

Same architecture. Vary the instruction-data fraction. Evaluate after one epoch.

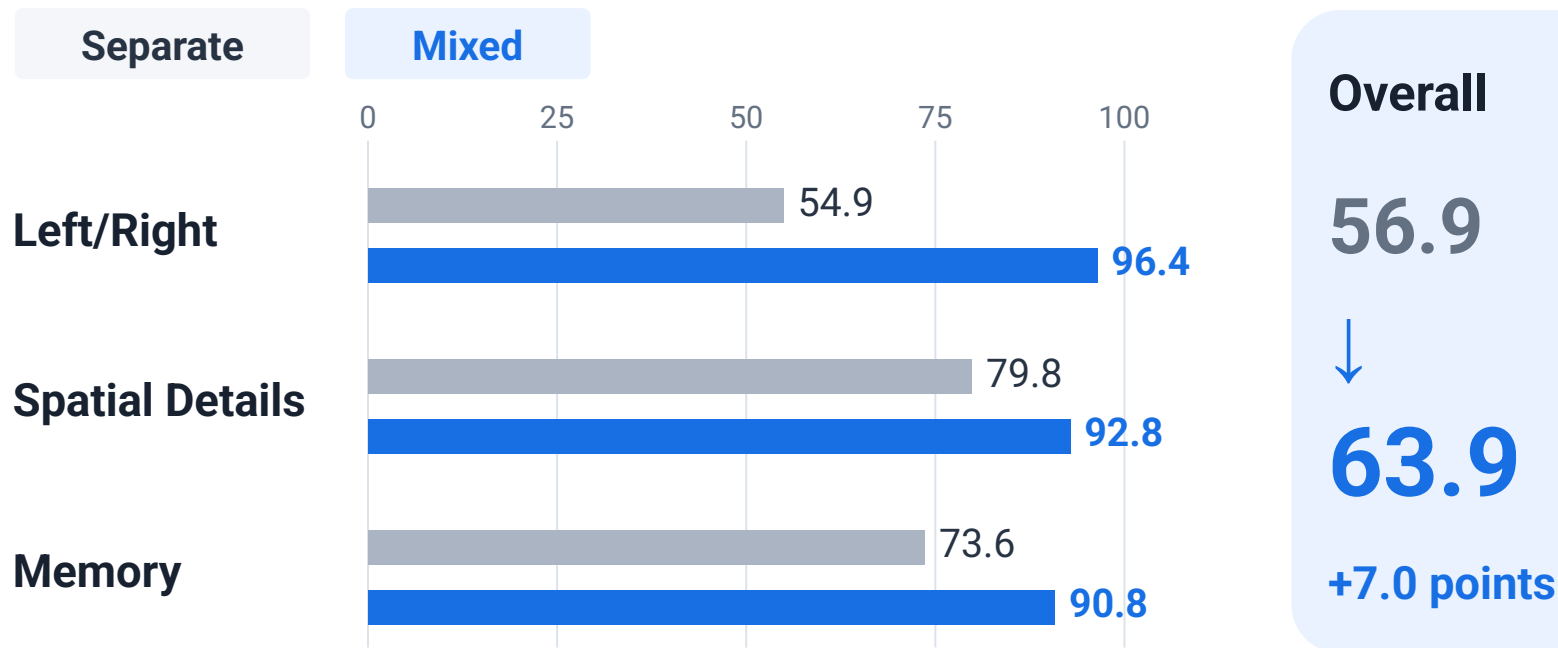


**Pretraining
wins at every
tested
fraction.**

Both models improve as more instruction data is supplied.

Task labels complement earlier experience; they do not replace it.

Ablation - Mixed-task SFT improves the overall score



Not every metric improves: naturalistic Who Has More, 62.1 → 60.5.

Ablation - Alignment and filtering matter

Change the language signal

Caregiver utterances



GPT-4o captions



Only +1.5 points overall

Change the filtering threshold

No filter  56.0

Default  63.9

Strict  67.3

Strict filtering keeps less than half the default pretraining data.

Limitation - Transfer across visual domains is limited

SAYCam → Ego4D

Same task construction;
adult-egocentric scenes.

	SAYCam	Ego4D
Overall	63.9 →	48.8
Spatial Details	92.8 →	47.1
Memory	90.8 →	72.4

Original NIH Toolbox

Small test sets; correct / total

Who Has More	13 / 24
Object Counting	2 / 6
Mullen Visual Reception	3 / 12

Limitation - Related tasks do not yet transfer

Seen during SFT

Picture Vocabulary



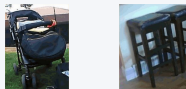
Object Counting



1 image
1–12 objects

Held out from SFT

Looking While Listening



4 choices → 2
choices

Subitizing

Blank

1–4

Blank

Reported
outcome

**Near
chance**

**Near
chance**

BabyVLM-v2: Future Work

Toward **Developmentally Plausible** Vision Foundation Models



BabyVLM-V2

Max Whitton

maxwh@bu.edu

Is touch a mechanism for human learning?

Making Sense of Touch from the Child's View for Contrastive Learning

Max Whitton, Zecheng Wang*, Puchen Liu*, Quang Tuan Truong, Shengao Wang, Manaswi Yadamreddy, Oktay Ozel, Visista Jayanti, Saniya Sekhon, Hanna Samuel Tadesse, Lawrence Miao, Junjie Wang, Jiasen Lu, Chen Yu¹, and Boqing Gong

Project Homepage: https://max-whitton.github.io/Contrastive_Learning_From_Touch

Abstract—Is the sense of touch a mechanism for human babies' learning of visual concepts? If so, can we quantify its importance, and to what extent do babies rely on their sense of touch for visual learning? To approach these questions in a principled way, we propose a structured coding system for baby-centric touch events, yielding a dataset of 264k two-second clips of touch events coded according to this system. Using this dataset, we pretrain developmentally grounded models that reveal promising insights into the nature of baby learning from touch.

input as we can from their real life, without adding any extra data that a baby would not have access to. However, it is prohibitively difficult to capture a baby's high-fidelity tactile sensations at scale directly. We approach this problem in a principled way, without introducing intrusive physical sensors, by leveraging a structured coding system for babies' touch events. We apply this taxonomy of touch to longitudinal, egocentric video recordings from three infants'

Oral Session 1 for more information!

Jun 2026

Announcement: The BabyVLM Challenge!



BabyVLM Challenge

Pretraining vision-language models on developmentally grounded data
inspired by infant cognition and learning.

[Submit Results](#)[Explore Pretraining](#)[Explore Task](#)

ABOUT

Overview

The BabyVLM Challenge is a shared task run in partnership with the [BabyLM Challenge](#), focused on developmentally plausible, sample-efficient vision-language modeling. Participants work with new datasets and evaluations purpose-built for small vision-language models trained on limited data.

The challenge will be introduced at the [BabyVLM workshop](#) with a tutorial. No prior BabyLM experience is needed to participate.



Pretraining

Pretrain on a compact dataset of egocentric audiovisual streams, provided below.



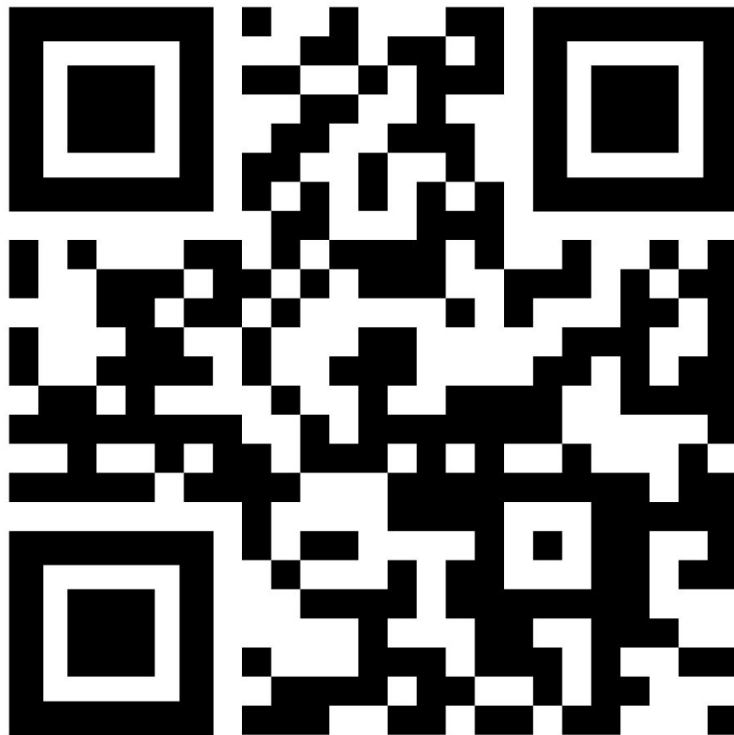
Task

Report scores on the 10 subtasks of the DevCV Toolbox.



Deadline

Coming soon!



Q&A



BabyVLM-V2

Until 12:50

How to formalize
developmentally plausible
learning? **BabyVLM-V2.**

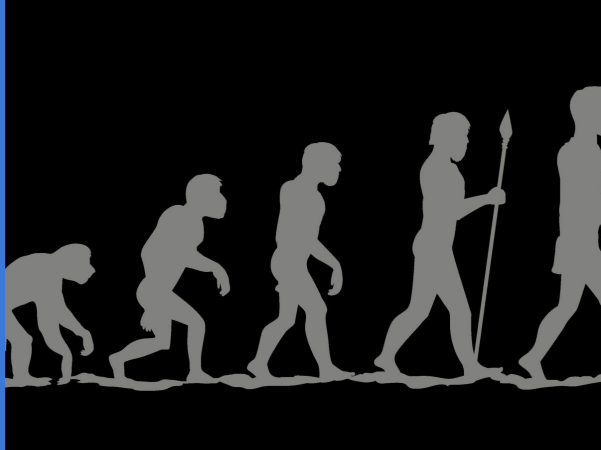
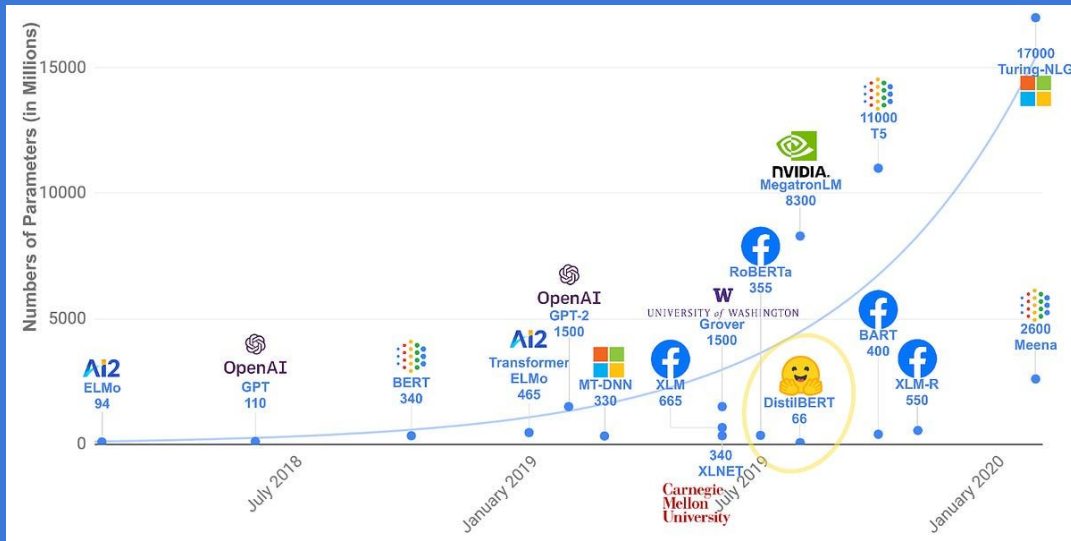
BabyVLM-V2: Toward Developmentally Grounded Pretraining and Benchmarking of Vision Foundation Models

Shengao Wang · Wenqi Wang · Zecheng Wang · Max Whitton · Michael Wakeham · Arjun Chandra · Joey Huang · Pengyue Zhu · Helen Chen · David Li · Jeffrey Li · Shawn Li · Andrew Zagula · Amy Zhao · Andrew Zhu · Sayaka Nakamura · Yuki Yamamoto · Jerry Yokono · Aaron Mueller · Bryan A. Plummer · Kate Saenko · Venkatesh Saligrama · Boqing Gong

What does **BabyVLM-V2** unlock?

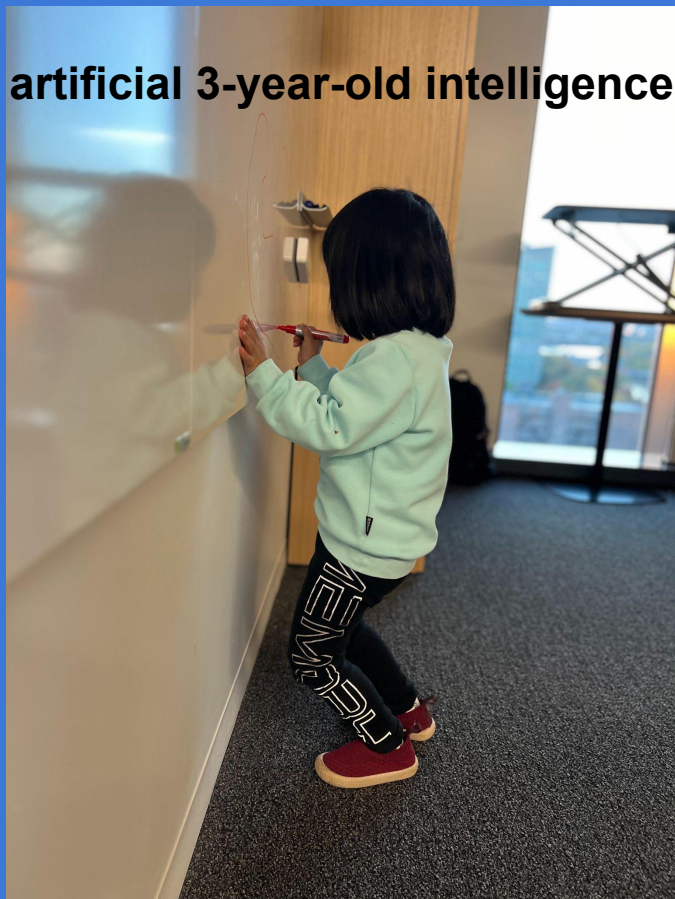
Making Sense of Touch from the Child's View for Contrastive Learning

Max Whitton, Zecheng Wang*, Puchen Liu*, Quang Tuan Truong, Shengao Wang, Manaswi Yadamreddy, Oktay Ozel, Visista Jayanti, Saniya Sekhon, Hanna Samuel Tadesse, Lawrence Miao, Junjie Wang, Jiasen Lu, Chen Yu¹, and Boqing Gong



- Two proven paths to foundation models
- Scaling law
 - **Human development**

artificial 3-year-old intelligence

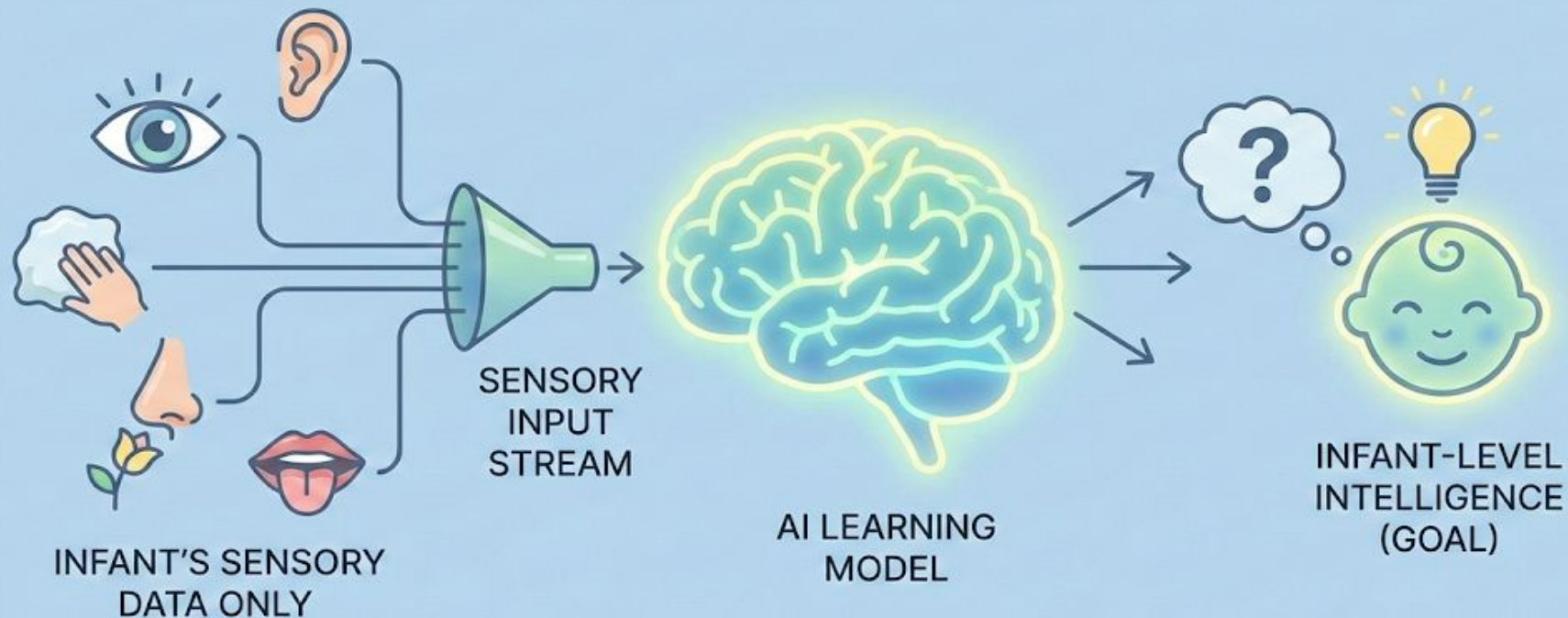


Two proven paths to foundation models

- Scaling law
- **Human development**

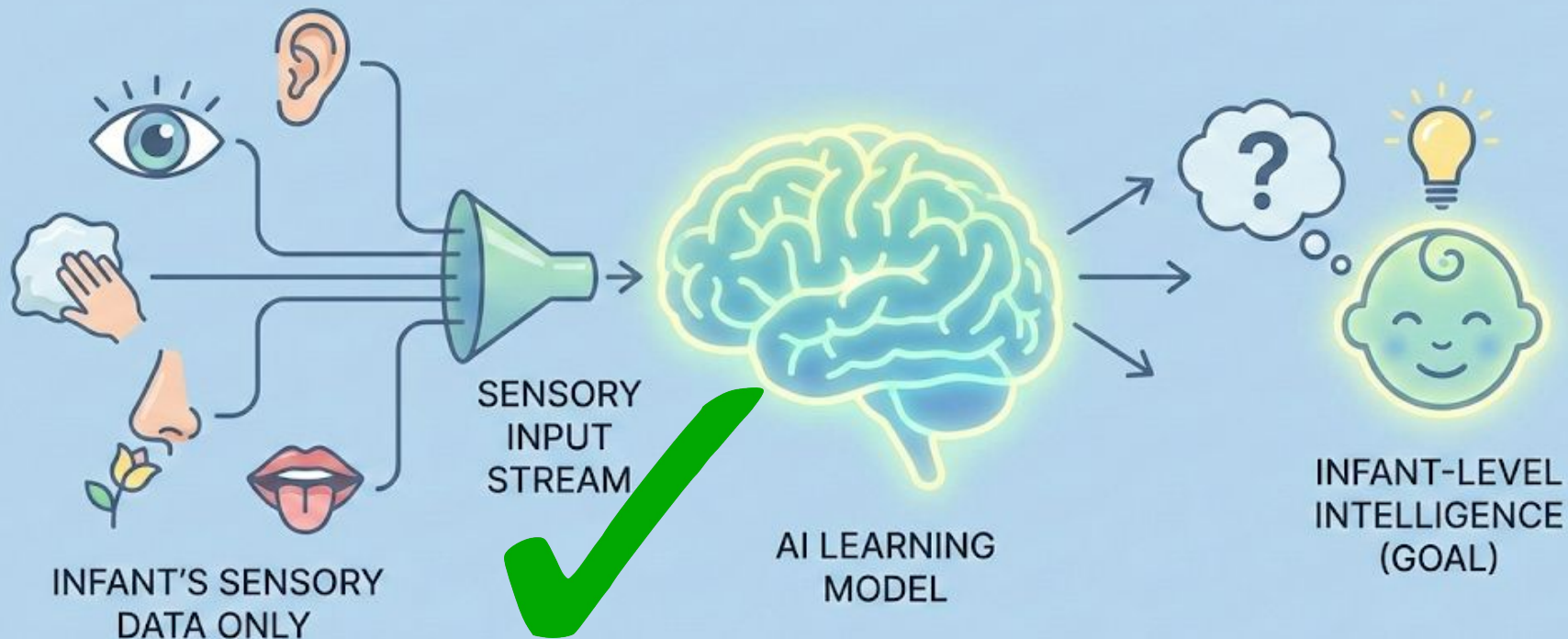


Can we achieve infant-level intelligence using infants' sensory data?





Can we achieve infant-level intelligence using infants' sensory data?



Benchmark validity & developmental fidelity

Table 4. **Performance comparison of different models on *DevCV Toolbox* (in-domain).** Different background colors denote different model families. We report accuracy (%) for all tasks; the higher, the better.

Model	Overall	Count	LeftRight	Spatial	PV	Memory	Localization	Visual Delay Response			Who Has More	
								binary	multi-exact	multi-adjacent	synthetic	naturalistic
Upper bound												
Human performance	93.0	99.1	94.5	100	91.8	97.9	87.3	98.2	63.6	95.5	98.2	96.4

- Adult performance data ***confirms the validity*** of our benchmark
- Adult participants reach ~100% when doubling their time on each task
- Extending this survey to ***children***

(Surprisingly?) Challenging for GPT-5 and Gemini-2.5-pro

Table 4. **Performance comparison of different models on *DevCV Toolbox* (in-domain).** Different background colors denote different model families. We report accuracy (%) for all tasks; the higher, the better.

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Proprietary models												
Gemini-2.5-flash	72.7	71.1	34.9	73.8	91.2	96.9	84.8	75.9	42.4	70.3	87.5	70.7
GPT-4o	74.6	39.0	89.8	92.6	93.7	99.7	81.7	64.2	29.3	62.9	87.9	79.3
Gemini-2.5-pro	82.5	77.2	68.8	90.5	93.8	97.8	88.8	86.9	54.0	87.7	90.6	71.7
GPT-5	87.6	69.1	96.0	94.5	95.0	99.9	85.2	95.1	62.9	90.1	88.9	86.6

- GPT can hardly count beyond 5**

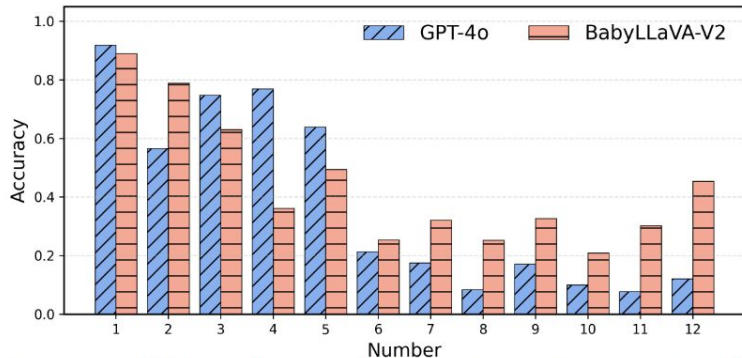
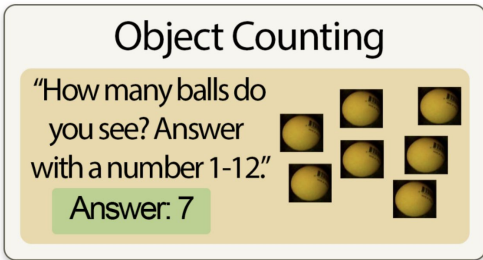


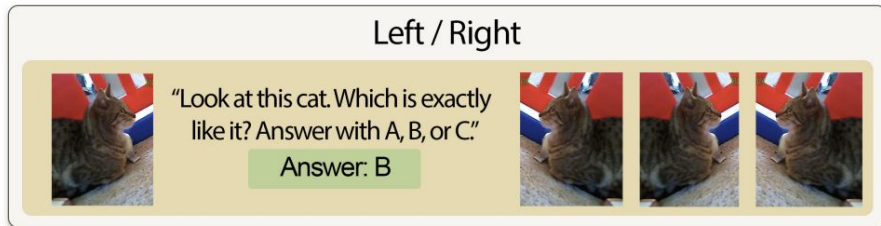
Figure 6. GPT-4o and our model's counting performance by different object numbers.

(Surprisingly?) Challenging for GPT-5 and Gemini-2.5-pro

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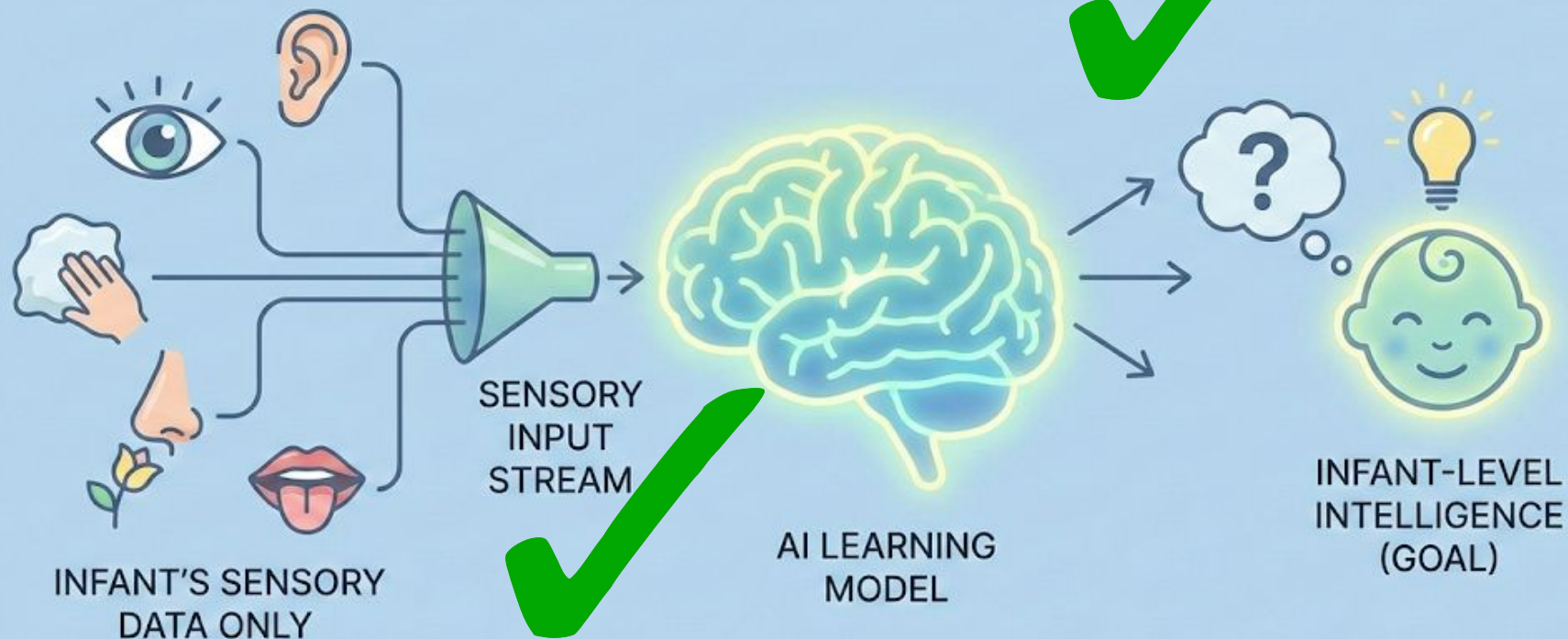
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GPT-5	87.6	69.1	96.0	94.5	95.0	99.9	85.2	95.1	62.9	90.1	88.9	86.6

- **GPT can hardly count beyond 5**
- **Gemini has left-right confusion**



Can we achieve infant-level intelligence using infants' sensory data?

How to assess the "infant-level intelligence"?



A baseline, encouraging baby model, pretrained from scratch

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Open-source models													
LLaVA-OneVision-0.5B	33.2	43.5	33.7	28.7	23.5	24.0	12.3	58.9	7.31	49.2	37.3	46.2	
InternVL3.5-1B	37.2	27.9	32.2	34.6	34.4	25.8	44.8	64.1	11.6	36.8	47.8	49.1	
Qwen2.5-VL-3B	47.0	29.2	33.7	40.0	71.7	36.5	85.8	66.7	17.0	32.7	51.7	52.3	
Baby models (Ours)													
BabyLLaVA-V2	55.2	44.6	42.3	91.3	27.4	75.3	38.8	57.6	33.1	45.6	98.4	52.8	
Lower bound													
Random guess	31.8	8.33	33.3	33.3	25.0	25.0	25.0	50.0	12.5	37.5	50.0	50.0	

- The baby model outperforms open-source models of similar sizes

A baseline, encouraging BabyLLaVA-V2, pretrained from scratch

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Qwen2.5-VL-3B	47.0	29.2	33.7	40.0	71.7	36.5	85.8	66.7	17.0	32.7	51.7	52.3	
Baby models (Ours)													
BabyLLaVA-V2	55.2	44.6	42.3	91.3	27.4	75.3	38.8	57.6	33.1	45.6	98.4	52.8	
Lower bound													
Random guess	31.8	8.33	33.3	33.3	25.0	25.0	25.0	50.0	12.5	37.5	50.0	50.0	

- The baby model outperforms open-source models of similar sizes
- The baby model beats GPT-4o at Count, Who Has More, & Visual Delay Response

BabyVLM-V2: Developmentally plausible pretraining & evaluation



Pretraining

Infants' **sensory** audio-video stream

- Longitudinal, egocentric
- Purposely & naturally restrictive



Evaluation

Tasks adapted from NIH Baby Toolbox

- Developmentally aligned
- Word-referent mapping, executive function, math



BabyVLM:

Leveraging psychology theory to answer questions about Machine Learning

A Sense of Touch:

Leveraging Machine Learning to answer questions about psychology

We have shown that speech benefits visual learning



You want to pour that in the **bucket**?



Want a bit more on your **spoon**?

Vong, W.K., Wang, W., Orhan, A.E., & Lake, B.M. (2024).
Grounded language acquisition through the eyes and ears of a single child. *Science*, 383(6682), 504–511.
<https://doi.org/10.1126/science.adi1374>

How much “Touch” can benefit visual learning?



Human Annotator: In a Play Setting, Holding/manipulating toy with your hand.

Generic: In a Play Setting, Holding/manipulating, Poking, toy with your hand.



Human Annotator: In a Eating Setting, Holding/manipulating an unknown object with your hand.

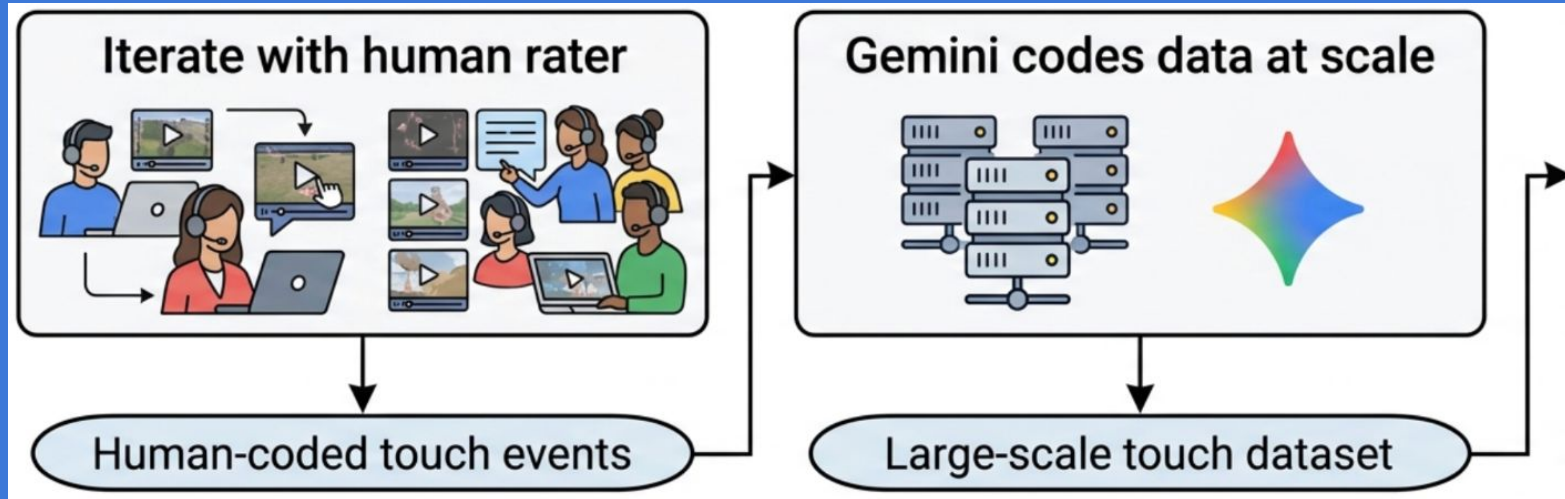
Generic: In a Eating Setting, Holding/manipulating, Grasping, food (scrambled eggs) with your hand.



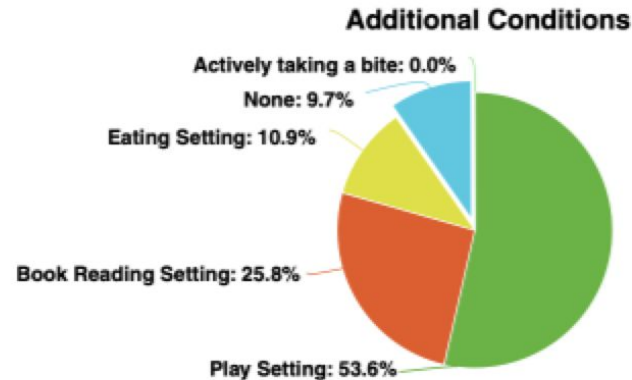
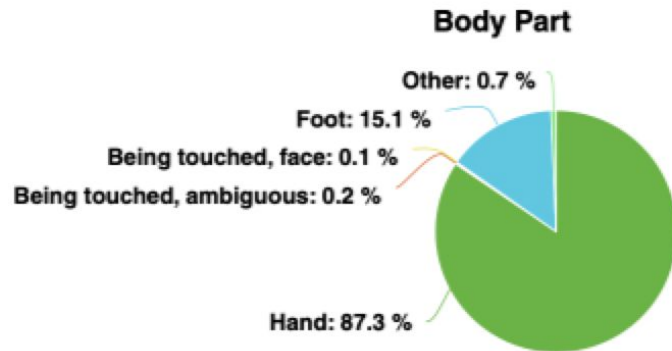
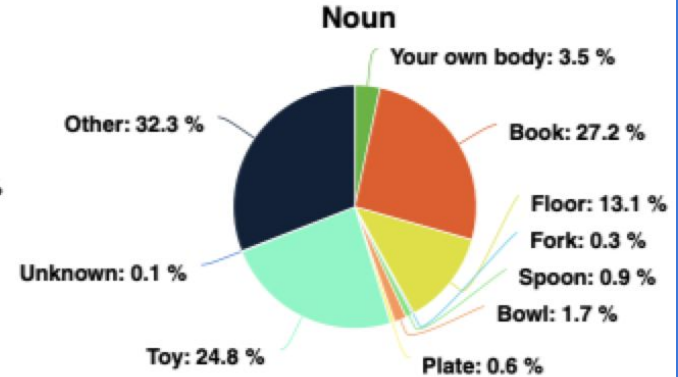
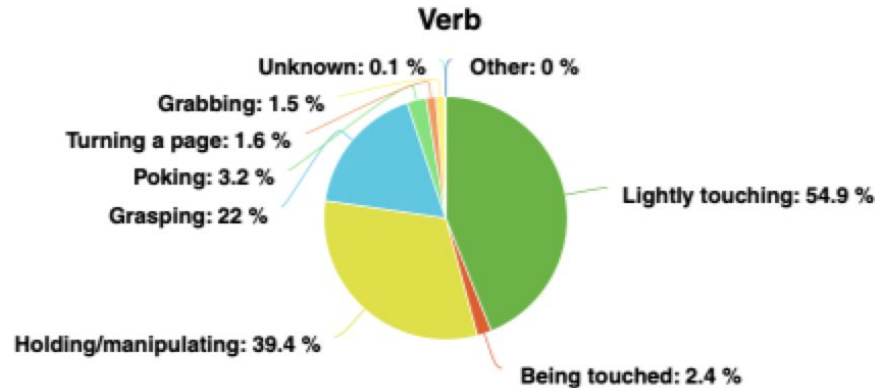
Human Annotator: In a Book Reading Setting, Turning a page of Book with your hand.

Generic: In a in a Book Reading Setting, Play Setting, Holding/manipulating Book with your hand.

Semi-automatic annotation of “Touch”



Result: a dataset of over 200,000 learning events!



Result: a dataset of over 200,000 learning events!

Verb

- Lightly touching
- Holding/Manipulating
- Grasping
- Poking
- Being touched
- Turning a page
- Grabbing
- ...

Noun

- Book
- Floor
- Your own body
- ...

Body part

- Hand
- Foot
- Face
- Ambiguous
- ...

Additional conditions

- Play setting
- Eating setting
- Book reading setting
- Actively taking a bite
- ...

Time locked with speech learning events!



Speech: Look at all those players!

Touch: [poking, book, hand]

Time locked with speech learning events!



Speech: Look at all those players!

~~**Touch:** [poking, book, hand]~~

Touch ID: 41

An important clarification



Speech: Look at all those players!

Touch: [poking, book, hand]

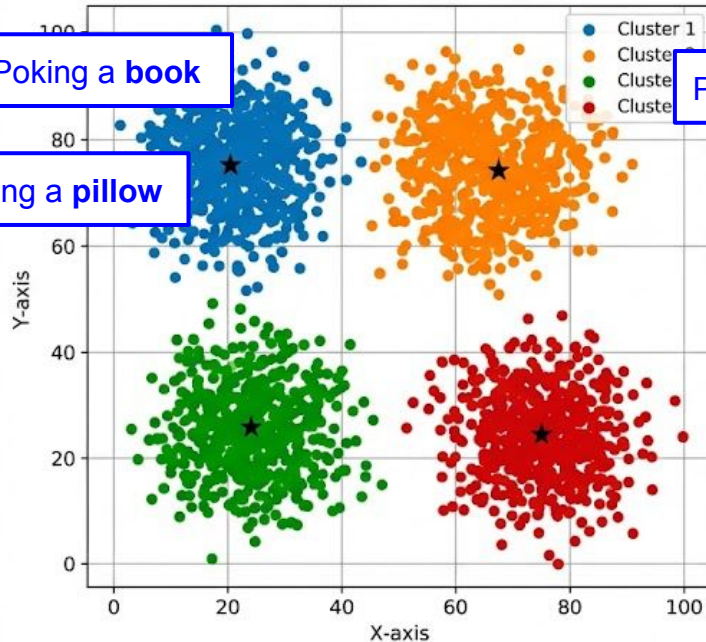
TouchID: 41

How many clusters?

K=4 (Four Clusters)

Poking a book

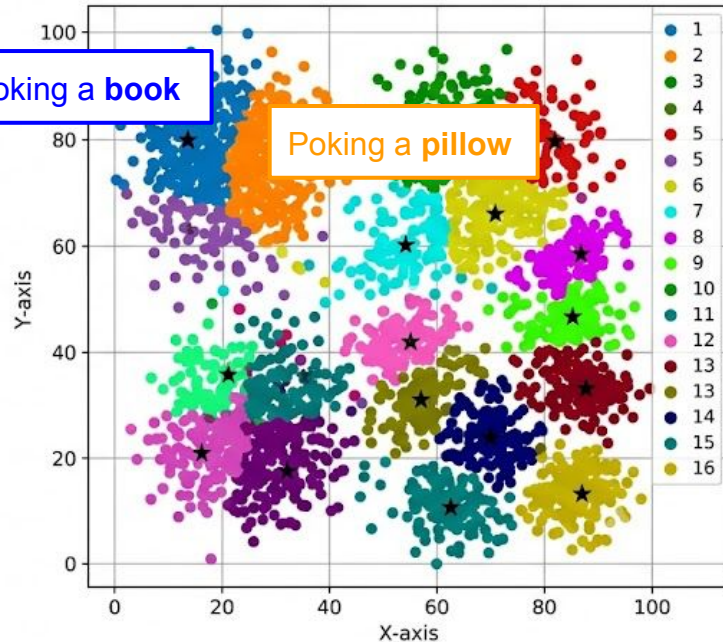
Poking a pillow



Poking a book

Poking a pillow

K=16 (Sixteen Clusters)



"Touch" enhanced CLIP



Contrastive Learning:

Optimize encoders to make these match



Optimize encoders to make these NOT match



Scene 1:



Vision Encoder

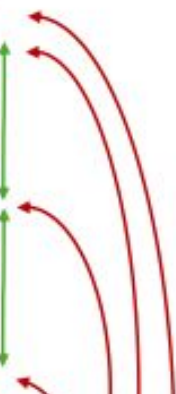


Touch Encoder



"Look at all those players!"

Text Encoder



Scene 2:



Vision Encoder



Touch Encoder



"Can you put the ball in the cup?"

Text Encoder

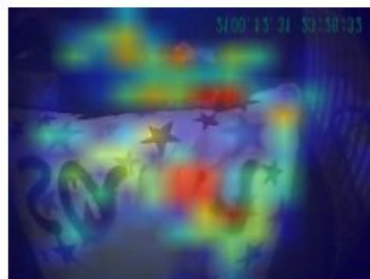
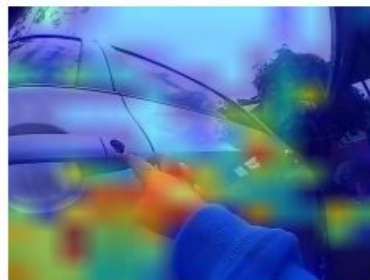
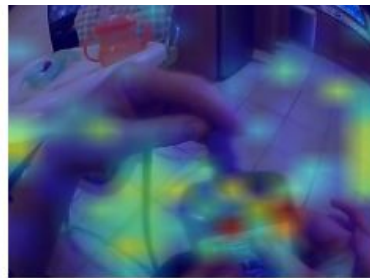


Qualitative results

Input Image

Touch Enhanced

Speech

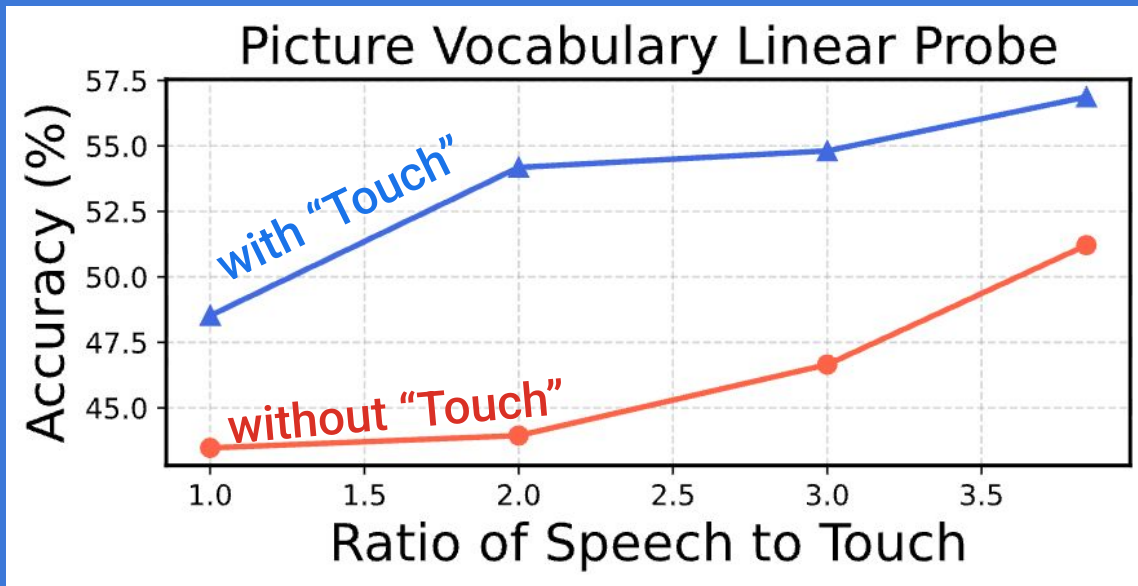


Fine grained evaluation

Which image is a **stroller**?



Fine grained evaluation



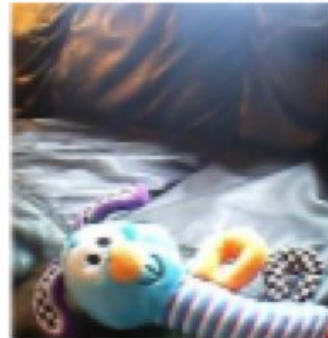
Fine grained evaluation

TABLE II: Evaluation performance for different numbers K of touch clusters

K	PV
16	54
64	57
256	57

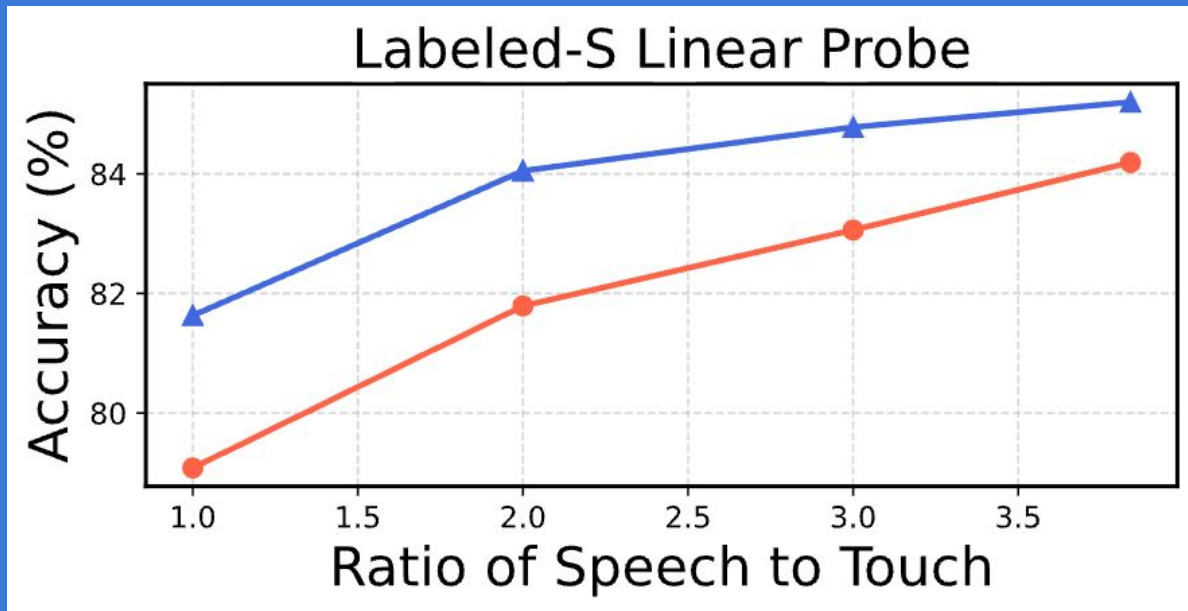
Coarse grained evaluation

Which image is a **cat**?



Vong, W.K., Wang, W., Orhan, A.E., & Lake, B.M. (2024).
Grounded language acquisition through the eyes and ears of a
single child. *Science*, 383(6682), 504–511.
<https://doi.org/10.1126/science.adi1374>

Coarse grained evaluation



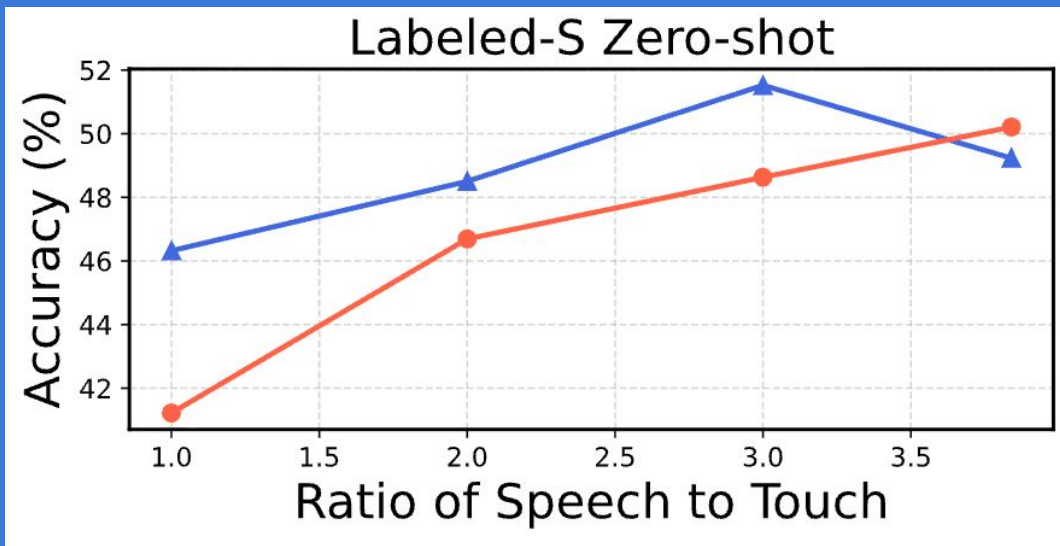
Coarse grained evaluation

TABLE II: Evaluation performance for different numbers K of touch clusters

K	Labeled-S (probe)	Labeled-S (zero-shot)
16	86	51
64	85	50
256	85	49

“Touch” helps visual learning

...especially when we artificially limit the quantity of speech data



Where does “Touch” NOT help?



Challenging Tasks

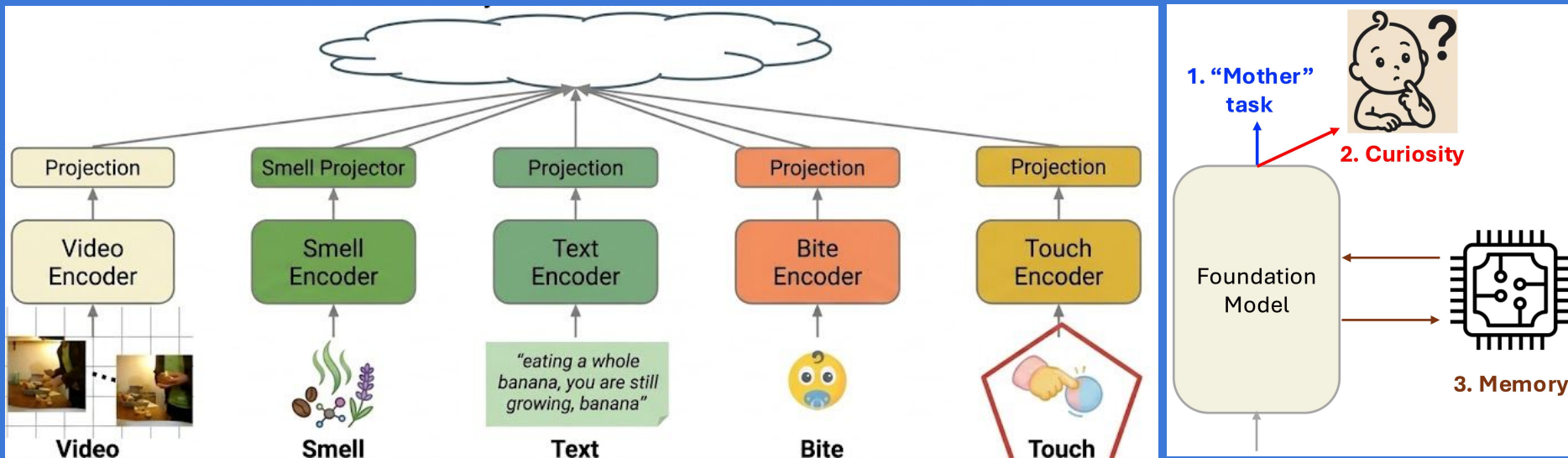
*Counting, localization,
who has more, etc.*



Simple Tasks

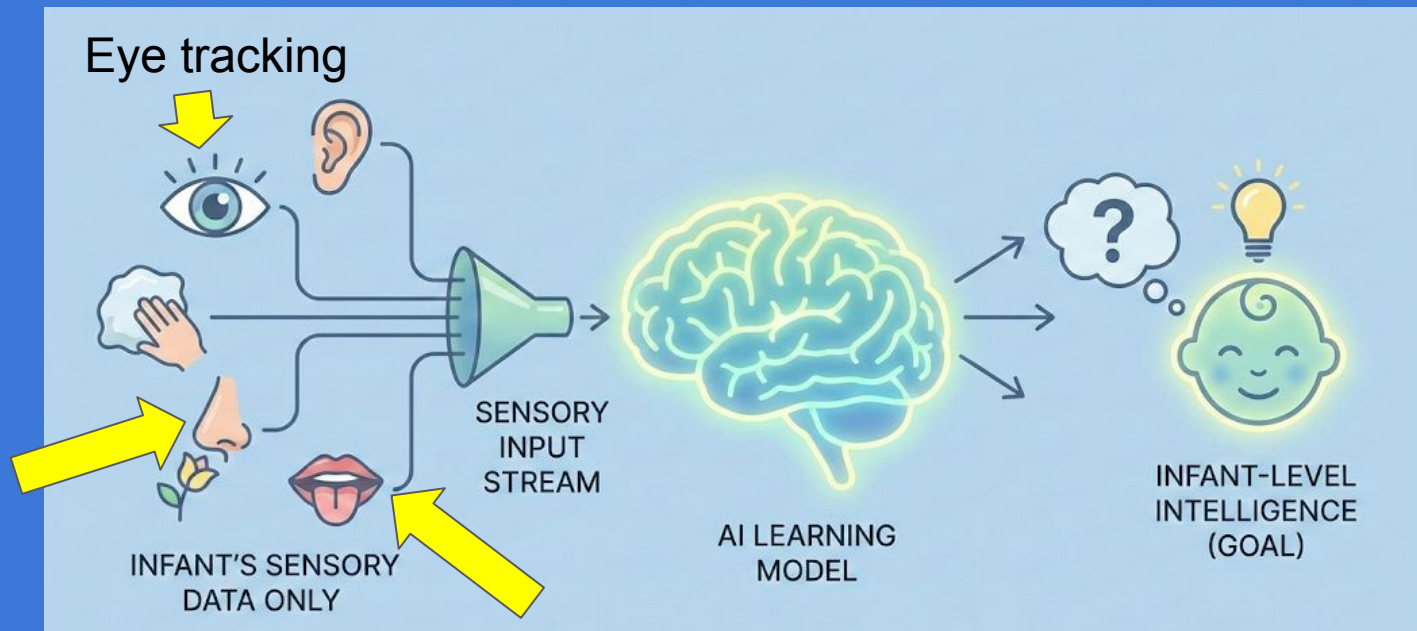
left/right, memory, etc.

Our wish list for BabyVLM-V2



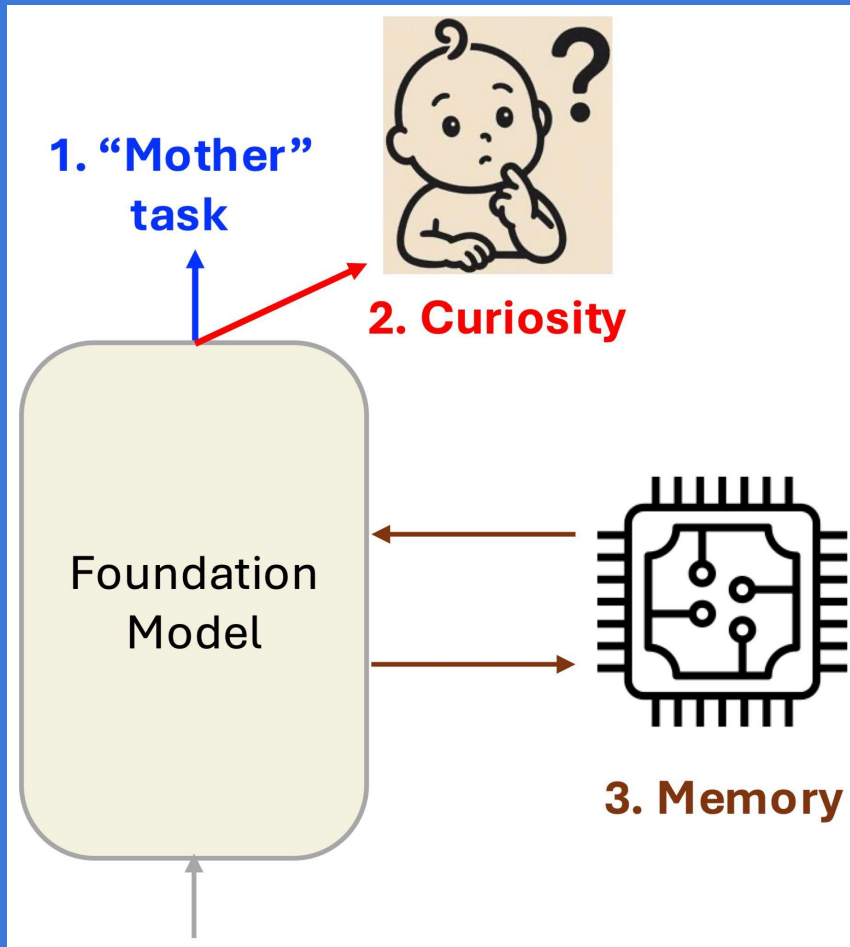
time-scheduled curriculum learning

Our wish list for **BabyVLM-V3**



time-scheduled curriculum learning

Our wish list for **BabyVLM-V3**



Our research agenda

- Ongoing human survey to get real child results on the DevCV Toolbox
- Building a large scale bite and smell dataset modeled after the methodology in *A Sense of Touch*
- Ongoing collaboration with Professor Chen Yu of UT Austin to collect in-the-wild eye tracking data
- Merging SAYCam with BabyView to approach the quantity of visual input a single baby sees during development
- Actively pursuing new collaborations to model a baby's learning process!

BabyVLM-V2 & Sense of Touch for Visual Learning

Can we achieve infant-level intelligence using infants' sensory data?



BabyVLM-V2



Sense of Touch

Boqing Gong

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Presented by

Arjun Chandra: ac25@bu.edu

Max Whitton: maxwh@bu.edu

Notes

Make motivation

Reanimate info about toolbox

Change order of sections